Physiome-ODE: A Benchmark for Irregularly Sampled Multivariate Time Series Forecasting Based on Biological ODEs

Abstract

State-of-the-art methods for forecasting irregularly sampled time series with missing values predominantly rely on just four datasets and a few small toy examples for evaluation. While ordinary differential equations (ODE) are the prevalent models in science and engineering, a baseline model that forecasts a constant value outperforms ODE-based models from the last five years on three of these existing datasets. This unintuitive finding hampers further research on ODE-based models, a more plausible model family. In this paper, we develop a methodology to generate irregularly sampled multivariate time series (IMTS) datasets from ordinary differential equations and to select challenging instances via rejection sampling. Using this methodology, we create `Physiome-ODE`, a large and sophisticated benchmark of IMTS datasets consisting of 50 individual datasets, derived from ODE models developed by research in Biology. `Physiome-ODE` is the first benchmark for IMTS forecasting that we are aware of and an order of magnitude larger than the current evaluation setting. Using `Physiome-ODE`, we show qualitatively completely different results than those derived from the current four datasets: on `Physiome-ODE` deep learning methods based on ODEs can play to their strength and our benchmark can differentiate in a meaningful way between different IMTS forecasting models. This way, we expect to give a new impulse to research on irregular time series modeling.

1 Introduction

Over the past decade, substantial research has been devoted to fully observed and regularly sampled time series, often referred to as regular (multivariate) time series. However, in certain scenarios, observations occur at irregular intervals and variables are measured independently resulting in a sparse multivariate time series with many missing values. The resulting time series are known as irregularly sampled multivariate time series with missing values, or simply **IMTS**.

With regular time series forecasting being a well-established topic, there exist a variety of sophisticated methods and standardized benchmarks (Godahewa et al., 2021; Gilpin, 2021; Bauer et al., 2021). On the other hand, there is currently no IMTS forecasting benchmark published. Creating such a benchmark demands a high number of publicly available time series datasets with genuinely irregular observation processes. However, only a few of those exist. Although they are closely related to each other, most evaluation sections of IMTS forecasting papers (Biloš et al., 2021; Schirmer et al., 2022; Yalavarthi et al., 2024; Zhang et al., 2024) are based on at least a subset of these datasets. Hence, it stands to reason that they induce a dataset bias. Additionally, we show that a simple baseline, which always predicts a constant value independent of time, is competitive with or even outperforms complex neural ODE models on these datasets. Hence, it appears questionable whether the currently used datasets are indeed well-suited for forecasting.

To address this limitation, we introduce `Physiome-ODE`, a wide benchmark of IMTS datasets consisting of 50 individual datasets, derived from ordinary differential equations from biological research, that are stored in the Physiome Model Repository (PMR). Biological processes are well-suited for generating IMTS datasets, as they are inherently multivariate and irregularly measured in real-world experiments. Additionally, the PMR provides Python implementations of many of these models, allowing us to create `Physiome-ODE` in an automated manner. While Biology researchers create their models based on very few and non-published observations, they enable us to create an arbitrary number of time series which relate to possible measurements of a real-world phenomenon. `Physiome-ODE` is the first benchmark for IMTS forecasting that we are aware of and an order of magnitude larger than the current evaluation setting of just four datasets.

Furthermore, to evaluate the complexity of the different forecasting datasets, we introduce a simple metric called *Joint Gradient Deviation (*JGD*)*, which measures the gradient variance of ODE solutions. We will show that our benchmark consists of datasets of different complexity and covers a wide range of different JGD values.

Finally, we evaluate current IMTS forecasting methods on `Physiome-ODE` and show that it includes many datasets on which neural ODE-based models significantly outperform the time-constant baseline model. Furthermore, a member of the neural ODE model family actually emerges as the overall most accurate model on `Physiome-ODE`. However, the datasets in `Physiome-ODE` are diverse enough that no single model is the most accurate for every dataset.

`Physiome-ODE` is a significant step forward to standardized IMTS forecasting and to monitor research progress. Our contributions include the following:

1. We introduce a simple baseline model that is restricted to making constant forecasts. On traditional IMTS forecasting evaluation, this simple baseline shows competitive or even better forecasting accuracy when compared to ODE-based models.
2. We propose *Joint Gradient Deviation (*JGD*)*, a simple score designed to approximate the difficulty of time series datasets for forecasting.
3. We create `Physiome-ODE`, a large and diverse benchmark of IMTS forecasting datasets. Maximizing JGD, we select and configure ODE models present in the Physiome (Yu et al., 2011) repository. `Physiome-ODE` is the first benchmark for IMTS forecasting that we are aware of.
4. We evaluate state-of-the-art IMTS forecasting models on `Physiome-ODE`. Our experiments show that the datasets included in `Physiome-ODE` are diverse enough to highlight different strengths of competing models. This results in different performance rankings of competing models for each dataset and the absence of a single best model. In that, `Physiome-ODE` differs from the currently used IMTS forecasting datasets, which is expected to provide a fresh impulse to research on IMTS forecasting models.
5. We share our code on GitHub
`Physiome-ODE`)

2 Irregularly Sampled Time Series Forecasting

An *irregularly sampled multivariate time series (IMTS)* is a sequence $X \coloneqq ((t_i, v_i))_{i=1:I} \in (\mathbb{R} \times (\mathbb{R} \cup \{\texttt{NaN}\})^C)^* \eqqcolon \mathcal{X}$ of pairs where t_i is the observation timepoint and v_i is an observation event with maximum of C-many variables (or channels) being observed. We use * to indicate the space of sequences. We call $v_{i,c} \in \mathbb{R}$ an observed value and $v_{i,c} = \texttt{NaN}$ a missing value. $Q \coloneqq (t_k^{\text{QRY}}, c_k^{\text{QRY}})_{k=1:K} \in (\mathbb{R} \times \{1, \ldots, C\})^* \eqqcolon \mathcal{Q}$ is the query sequence where t_k^{QRY} is a queried (future) timepoint and c_k^{QRY} is a queried channel. $Y \coloneqq (y_1, \ldots, y_K) \in \mathbb{R}^* \eqqcolon \mathcal{Y}$ is the forecasting answer where y_k is the ground truth forecast value for Q_k.

Then the *IMTS forecasting problem* is defined as follows: given a sequence of triples $((X_n, Q_n, Y_n))_{n=1:N}$ drawn from a random distribution ρ where $X_n \in \mathcal{X}$ is an irregularly sampled

Table 1: Test MSE for forecasting next 50% after 50% observation time. OOM refers to out of memory. We highlight the best model in **bold** and underline the second best. † indicates that we show the results from Yalavarthi et al. (2024)

Model	USHCN	PhysioNet-2012	MIMIC-III	MIMIC-IV
GRU-ODE	1.017±0.325	0.653±0.023†	0.653±0.023†	0.439±0.003†
LinODEnet	0.662±0.126	0.411±0.001†	0.531 ± 0.022†	0.336±0.002†
CRU	0.730±0.264	0.467±0.002†	0.619±0.028†	OOM†
Neural Flow	1.014±0.336	0.506±0.002†	0.651±0.017†	0.465±0.003†
GraFITi	**0.636±0.161**	**0.401±0.001**†	**0.491 ± 0.014**†	**0.285±0.002**†
GraFITi-C	0.875±0.204	0.407±0.001	0.543±0.024	0.324±0.002

time series, $Q_n \in \mathcal{Q}$ is a query sequence and $Y_n \in \mathcal{Y}$ is a ground truth answer, as well as a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$, e.g., the mean squared error between two equilength sequences, the task is to find a model $\hat{Y} : \mathcal{X} \times \mathcal{Q} \to \mathcal{Y}$ such that the expected loss between the ground-truth forecasting answer Y and the model prediction $\hat{Y}(X, Q)$ is minimal:

$$\mathcal{L}(\hat{Y}; \rho) \coloneqq \mathbb{E}_{(X,Q,Y)\sim\rho}\, \ell(Y, \hat{Y}(X, Q)). \tag{1}$$

2.1 Current Forecasting Models

In our experiments we compare five recent models. Four of these five network architecture are variants of the neural ODE (Chen et al., 2018), which uses an ODE, that is defined by a neural network, to infer the hidden state in-between observations. At observation times the neural ODE-based estimation of the hidden state is updated on the actual observation. For more details, we refer to the papers. We use the following models:

GRU-ODE-Bayes (De Brouwer et al., 2019) utilizes a continuous version of Gated Recurrent Units (GRU) as the ODE defining network f. Additionally, GRU-Bayes applies a GRU-based Bayesian update to the hidden state at observations. **LinODEnet** (Scholz et al., 2022) restricts f to be linear functions but encodes data non-linearly into latent space. At observations LinODEnet updates its hidden state with a so-called nonlinear KalmanCell, a module inspired by Kalman Filtering (Kalman, 1960). **Continuous Recurrent Units** (Schirmer et al., 2022) replaces the neural ODE with a stochastic differential equation (SDE). CRU is able to infer the hidden state at any time in closed form using continuous-discrete Kalman filtering, instead of solving a neural ODE. **Neural Flows** (Biloš et al., 2021) infers the solution curve of an ODE directly with invertible neural networks. **GraFITi** (Yalavarthi et al., 2024) is substantially different from neural ODE. It encodes the observations from IMTS into a graph structure and infers forecasts by solving a graph completion problem with a graph attention network. On the established evaluation datasets GraFITi significantly outperforms the neural ODE-based models in terms of forecasting accuracy.

3 Time-Constant Approaches Outperform ODE-based Approaches in the Current Evaluation Scenario

Current Evaluation Scenario. Traditionally, IMTS forecasting has relied on datasets like PhysioNet2012 (Silva et al., 2012), MIMIC-III (Johnson et al., 2016), and MIMIC-IV (Johnson et al., 2021) obtained from Kaggle challenges designed for IMTS classification. Additionally, De Brouwer et al. (2019) proposed to create an IMTS forecasting task, based on USHCN (Menne et al., 2016), a fully observed regularly sampled time series dataset containing climate data. We present characteristics of these datasets in Table 3 (appendix).

The GraFITi model is the state-of-the-art on PhysioNet2012, MIMIC-III, MIMIC-IV, and USHCN, demonstrating a significant performance advantage over neural ODE-based methods (Yalavarthi et al., 2024). To further analyze this gap, we designed a *constant* version of GraFITi (GraFITi-C), which is restricted to make the exact same forecast for every query time point. This is achieved by training a GraFITi model that cannot access the actual query time point (t_k^{QRY}), but is instead always provided with a constant dummy query time point.

The results shown in Table 1 indicate, that neural ODE-based models are indeed outperformed by this allegedly easy-to-beat baseline on PhysioNet2012 and MIMIC-IV. On MIMIC-III the best performing ODE-based model achieves a lower average test MSE than GraFITi-C, however the difference is within the standard deviation. GraFITi on the other hand, consistently outperforms its constant variant. However, it also cannot separate in by a drastic margin. These findings could relate to important covariants, which are not contained in the dataset. For instance, it is impossible to predict a patient's blood glucose level without information about any food intake.

We designed `Physiome-ODE` to be an extension to the currently used evaluation protocol. For our ODE generated datasets, we can guarantee that all necessary covariants are given. Furthermore, we can take full control about the simulated observation process.

4 GENERATING CHALLENGING IMTS DATASETS FROM ODEs

Sampling ODEs to Generate IMTS Data. In science and engineering, processes that evolve in continuous time are usually described by differential equations, in the simpler cases by ordinary differential equations (ODEs). All possible observations of such a process with C **channels** over **duration** T at any time t can be captured by a time-varying function $x\colon [0,T] \to \mathbb{R}^C$ as $x(t)_c$. An ODE characterizes such a function implicitly by two conditions:

1. A relation between observation time t, observation value $x(t)$ and the instantaneous change in observation values $x'(t)$, i.e. the first derivative of x. This relation is described by a function $F : [0,T] \times \mathbb{R}^C \times \mathbb{R}^C \to \mathbb{R}$ called the **system**.
2. The **initial values** $x_0 \in \mathbb{R}^C$ of the function at time $t = 0$:

$$F(t, x(t), x'(t)) = 0 \quad \forall t \in [0,T], \qquad x(0) = x_0 \tag{2}$$

Note that ODEs can be defined including higher order derivatives, but w.l.o.g. one can assume they are of first order, since the higher order system can be transformed into a first order system by a change of variables (Teschl, 2012). In a few special cases the solution $x(t)$ can be computed analytically. For example, the one-dimensional system $F(t, x(t), x'(t)) := ax(t) - x'(t)$ characterizes the exponential growth function $x(t) = e^{at}x_0$. However, in most cases, it can only be computed by a numerical ODE solver.

Specific parameters that occur in a system F like the growth factor a in our example are called **constants** and one writes $F(t, x(t), x'(t); a)$ with $a \in \mathbb{R}^A$ to denote a system with A-many constants. Triples (F, a, x_0) of systems, constants and initial values can be turned into a simulator, i.e. a generative model, by complementing them with two more components:

1. A **sampling process** s for observation time points and channels that can generate finite sequences $((t_1, c_1), \ldots, (t_I, c_I))$ of pairs of observation times $t_i \in [0,T]$ and observed channels $c_i \in \{1, \ldots, C\}$.
2. A **noise model** p_{NOISE} for the observation values, that is a conditional distribution $p_{\text{NOISE}}(x^{\text{OBS}} \mid x^{\text{TRUE}})$ of the observed value given the ground truth value.

Then one can generate irregular sampled instances $(t_i, c_i, x_i^{\text{OBS}})_{i=1:I}$ of the (only implicitly given) underlying function x with missing values and affected by observation noise simply via:

$$\big((t_i, c_i)\big)_{i=1:I} \sim s \qquad x_i^{\text{TRUE}} := \text{ode-solve}(F, a, x_0, t_i)_{c_i} \qquad x_i^{\text{OBS}} \sim p_{\text{NOISE}}(x_i^{\text{OBS}} \mid x_i^{\text{TRUE}}) \tag{3}$$

Identifying Challenging Time Series. While any ODE implicitly describes a time-varying function $x\colon [0,T] \to \mathbb{R}$, not all such functions are challenging to forecast: For example, some just describe very simple convergence processes that approach a limit value over a long time. In order to discover challenging time series, we propose to measure the *mean gradient deviation* (MGD), that is the time-normalized L^2 norm of $x'(t)$ around its mean:

$$\text{MGD}(x) := \min_c \sqrt{\frac{1}{T}\int_0^T (x'(t) - c)^2 \, \mathrm{d}t} \tag{4}$$

Note that the minimum is obtained precisely for the mean gradient

$$c = \frac{1}{T}\int_0^T x'(t)\,\mathrm{d}t = \frac{x(T) - x(0)}{T}. \tag{5}$$

For a linear test function $x(t) = at + b$, the MGD is zero, while for a sine wave $x(t) = \sin(\omega t)$, the MGD scales linearly with the frequency ω. To identify not just individually challenging time series, but whole distributions (i.e. a generator p), one can compute the expectation of MGD. We found that the MGD grows super-exponentially with the Lipschitz constant, which is described in more detail in Appendix C

With MGD one does assess only the heterogeneity within each function, not heterogeneity between different functions, which is an important precondition for an interesting data set for machine learning tasks. To capture this second aspect, we propose the *mean point-wise gradient deviation* (MPGD):

$$\mathrm{MPGD}(p) := \frac{1}{T}\int_0^T \mathrm{std}_{x\sim p}[x'(t)]\,\mathrm{d}t = \frac{1}{T}\int_0^T \sqrt{\mathbb{E}_{x\sim p}\left[\left(x'(t) - \mathbb{E}_{y\sim p}[y'(t)]\right)^2\right]}\,\mathrm{d}t \tag{6}$$

We combine both measures multiplicatively as *joint gradient deviation* (JGD) for distributions of functions, i.e. function generators:

$$\mathrm{JGD}(p) := \mathrm{MPGD}(p)\cdot\mathbb{E}_{x\sim p}[\mathrm{MGD}(x)] \tag{7}$$

Note, that we can approximate MGD with a finite number of samples:

Lemma 1. *For a function $x \in \mathcal{C}^1([0,T])$ and $\epsilon > 0$ such that $\frac{T}{\epsilon} \in \mathbb{N}$ we consider the divided differences*

$$x_t^{\mathrm{diff},\epsilon} := \frac{x(t) - x(t-\epsilon)}{\epsilon}, \quad t = \epsilon, 2\epsilon, \ldots, T.$$

It then holds that the numerical estimator $\widehat{\mathrm{std}}[(x_k)_{k=1:K}] := \sqrt{\frac{1}{K}\sum_{k=1}^K (x_k - \bar{x})^2}$ of the standard deviation of the divided differences converges to the MGD *of the function:*

$$\widehat{\mathrm{std}}[x^{\mathrm{diff},\epsilon}] \xrightarrow{\epsilon\to 0} \mathrm{MGD}(x) \tag{8}$$

We can also approximate MPGD with a finite amount of samples under assumptions we explain in Appendix B.

Lemma 2. *The mean point-wise gradient deviation of a distribution p of functions can be approximated on a sample $(x_n)_{n=1:N}$ of N-many sequence from p on a fixed grid of time points as follows: Given the divided differences:*

$$x_{n,t}^{\mathrm{diff}} := \frac{x_n(t) - x_n(t-\epsilon)}{\epsilon}, \quad t = \epsilon, 2\epsilon, \ldots, T$$

Then the average of the point-wise std. of the divided differences converges to the MPGD*:*

$$\frac{\epsilon}{T}\sum_{t=\epsilon}^{T} \widehat{\mathrm{std}}[(x_{n,t}^{\mathit{diff}})_{n=1:N}] \longrightarrow \mathrm{MPGD}(p) \quad \textit{for} \quad \epsilon \to 0 \quad \textit{and} \quad N \to \infty \quad \textit{almost surely} \tag{9}$$

The proofs can be found in Appendix B. Due to Lemma 1 and Lemma 2, we can approximate the MGD, MPGD and therefore the JGD based on the set of evenly spaced sequences $X \in \mathbb{R}^{N\times T}$, where $x_{n,t}$ denotes the $value$ at time-step t in the n-th sequence, with $x_n \sim p$. Given $x_{n,t}^{\mathrm{diff}} := x_{n,t+1} - x_{n,t}$, it holds that $\mathrm{JGD}(p) \approx \mathrm{MPGD}(p)\cdot\frac{1}{N}\sum_{n=1}^N \mathrm{MGD}(x_n)$ and furthermore:

$$\mathrm{MGD}(x_n) \approx \widehat{\mathrm{std}}[x_n^{\mathrm{diff}}] \qquad\qquad \mathrm{MPGD}(p) \approx \frac{\epsilon}{T}\sum_{t=\epsilon}^{T}\widehat{\mathrm{std}}[(x_{n,t}^{\mathrm{diff}})_{n=1:N}] \tag{10}$$

Lemma 1 and Lemma 2 thus allow us to approximate the JGD-score by sampling the values of several functions on a shared grid from a generator. Its definition on the full function guarantees that we will always get the same results (in the bounds of the approximation error).

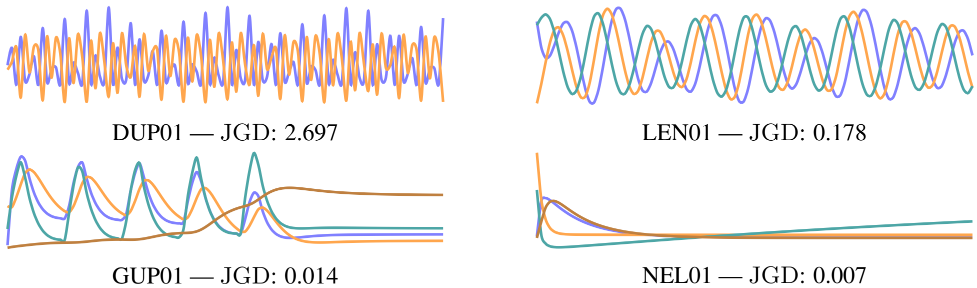

Figure 1: Demonstration of time series realized by 4 ODEs of different prediction difficulties without adding any noise. Each line / color represents a channel. Trajectories are shown for a duration of the respective σ_{dur} as shown in Table 4.

JGD for ODE models. To simplify the initial definition, we considered the JGD for a univariate function $x\colon [0, T] \to \mathbb{R}$, and we interpret a single channel within a multivariate ODE system as such a function. Computing the JGD for a multivariate ODE model requires combining the JGD values of each channel. We found that simply averaging all channels disadvantages models with a high number of channels. To address this, we compute the mean of the ten channels with the highest JGD. For models with ten or fewer channels, we revert to using the mean of all channels. Note that JGD is only a useful measure when applied to standardized data, otherwise the units are incomparable. Consequently, we normalize each channel to have a mean of 0 and standard deviation of 1 over all time steps and time series instances.

Varying Single ODE Instances. From the scientific literature one often gets single ODE instances: a triplet $(F, a^{\text{LIT}}, x_0^{\text{LIT}})$ of a parametrized ODE system F, its parameters/constants a^{LIT} and the initial conditions x_0^{LIT}, all three describing some specific experiment. To generate data by sampling only from such a single ODE instance will create functions that differ only in observation time points and channels and in the noise. These instances are too similar to each other, without noise there is only one function, and hence the MPGD of the distribution would be zero. We therefore propose to also carefully vary (i) the initial conditions x_0, (ii) the ODE constants a and (iii) the total duration T and of the ODE sampling process. We sample each from a distribution that yields the values given in the scientific literature on expectation and is controlled by a spread parameter σ each:

$$x_0 \sim p_{\text{initial}}(\,\cdot \mid x_0^{\text{LIT}}, \sigma_{\text{initial}}) \qquad a \sim p_{\text{const}}(\,\cdot \mid a^{\text{LIT}}, \sigma_{\text{const}}) \qquad T \sim p_{\text{dur}}(\,\cdot \mid \sigma_{\text{dur}}) \tag{11}$$

Let p be the two-stage function generator that first samples tuples (x_0, a, T) from their respective distributions and then yields as function x the solution to the ODE F with initial values x_0, constants a and running for a duration of T. Instead of blindly choosing the spreads σ, we optimize them w.r.t. the JGD-value of the generator:

$$(\sigma^*_{\text{initial}}, \sigma^*_{\text{const}}, \sigma^*_{\text{dur}}) := \underset{\sigma_{\text{initial}}, \sigma_{\text{const}}, \sigma_{\text{dur}}}{\arg\max} \ \text{JGD}\big(p(\,\cdot \mid \sigma_{\text{initial}}, \sigma_{\text{const}}, \sigma_{\text{dur}})\big) \tag{12}$$

The JGD of each generator is estimated by 100 randomly sampled fully observed time series with a sequence length of 100. Since we are interested in the difficulty of the forecasting task, we compute the JGD on the final 50 steps of sequence only. We search σ_{initial} from the range of $\{0.1, 0.3, 0.5\}$, σ_{const} from $\{0.05, 0.1, 0.3\}$, and σ_{dur} from $\{0.33, 1, 3.3, 10, 30\}$. The unit of σ_{dur} varies for each ODE model, it is given by Physiome (Yu et al., 2011) and listed in Table 4 (appendix).

Creating the `Physiome-ODE` Benchmark. In the Physiome database we identified $N = 208$ multivariate ODE models $(F^n, a^{\text{LIT},n}, x_0^{\text{LIT},n})_{n=1:N}$, that are defined by automatically generated Python code. We ranked all ODE-models with according to their optimized JGD-value (see Table 4) and selected the highest 50 datasets for `Physiome-ODE`. For each of the final datasets, we created a dataset with 2000 instances each. To further increase the range of initial states, we create datasets of lengths σ^*_{dur}, but with 200 observation steps. For each time series instance, we sample the initial time step $t_{\text{onset}} \sim \{t_i\}_{1 \le i \le 100}$ and then select the next 100 steps for each time series instance. Consequently, each time series instance has a duration of $\frac{1}{2}\sigma^*_{\text{dur}}$. While the time series instances are sampled regularly at first, we create IMTS instances by randomly dropping observations with a chance of 80%. For p_{noise} we select to add $\epsilon \sim \mathcal{N}(0, 0.05)$. In Figure 1 we show four trajectories contained in `Physiome-ODE` with the respective JGD.

Handling exploding ODEs. Varying the constants and initial states of the ODEs can lead to inconsistent states, which could never occur in real world scenarios. In general, this is not an issue regarding the machine learning perspective on our experiments. Occasionally however, certain combinations of inconsistent states and constants can lead to extreme values which are magnitudes greater than values that occur in other time series instances. These extreme values will dominate the training and evaluation loss in a problematic manner. We prevent this by rejecting all $(\sigma_{\text{initial}}, \sigma_{\text{const}}, \sigma_{\text{dur}})$ containing any values greater than 10 times the channel-wise standard deviation in the 100 samples created for evaluation. If such values nevertheless occur within the final 2000 samples created for `Physiome-ODE`, we drop the respective instances. We follow the same protocol to handle errors thrown by the ode-solver.

5 Experiments

Baseline models. We compare on 5 recent IMTS forecasting models, namely **GRU-ODE-Bayes** [1] (De Brouwer et al., 2019), **Neural Flows** (Biloš et al., 2021), Continuous Recurrent Unit (**CRU**; Schirmer et al., 2022), **LinODEnet** (Scholz et al., 2022; Mione et al., 2024) and **GraFITi** (Yalavarthi et al., 2024). GRU-ODE-Bayes, Neural Flows, CRU and LinODEnet are differential equations based models whereas GraFITi is a graph based model. We also introduce a naive model called GraFITi-Constant (GraFITi-C) which predicts a fixed value for all the future time points in all the target channels for an instance.

Experimental Protocol. In our experiments models have to predict the last 50% of the time series after observing the first 50%. We use 5-fold cross validation, and for each fold we split the data into training, validation and test set with a ratio of 70:20:10. Additionally, we resample the sparse observation mask to transform each instance into an IMTS. The evaluation metric is the mean square error (MSE). For simplicity, we randomly sample 10 hyperparameter configurations and fitted each model on a single fold per dataset, selecting the configuration with the lowest validation MSE. The winning configuration per model is then trained and evaluated on all 5 folds (see Appendix F). We run the benchmark experiments on a cluster of 4 NVIDIA 4090 GPUs with 24 GB.

Empirical Results. The numbers presented in Table 2 are the mean and standard deviation of MSE for 5 folds. Overall, LinODEnet (Scholz et al., 2022; Mione et al., 2024) is the best-performing model, winning the most datasets and having the highest average rank. On the other hand, the average rank of GraFITi (2.20) is close to that of LinODEnet (2.10). Surprisingly, GraFITi-C performs better than Neural Flows and GRU-ODE-Bayes and has an average rank of 2.86. This is because datasets like `DUP01` and `DOK01` are too hard to forecast, hence the complex models cannot improve upon the accuracy of a constant model. Other datasets are too simple that even a constant model yields MSE close to 0.0025, which is the lowest value possible to achieve due to the normal-distributed noise with a variance of 0.05.

In order to check how good our JGD score can measure the difficulty in forecasting of a time series we plot JGD score vs the best MSE score achieved in Figure 2. It can be observed that JGD fulfills its purpose of finding challenging and interesting ODE models. More specifically JGD and the test MSE of the best performing model are correlated with a Spearman coefficient of 0.77.

6 Extensions and Limitations

Currently, most datasets in `Physiome-ODE` still show a good potential for improvement: even the best performing methods are far away from the Bayes error of 0.0025 MSE. In case authors of future methods get the impression that their method could benefit from larger datasets, i.e., more instances per dataset than 2000, they easily can generate larger ones with 10,000, 100,000, etc. instances.

`Physiome-ODE` is not based on observation data as in these areas observed data usually is expensive to get, and thus only limited amounts exists or are publically available, at least. In fact, we scanned through the respective papers of our top 50 ODE models and could not find the original experimental data for a single one. Hence, if one wants to conduct machine learning experiments on these complex and rich biological processes one is forced to create them using the ODE models describing the process.

[1] For brevity, we denote this as GRU-ODE in the tables.

Table 2: Experimental results on various baseline models. `Physiome-ODE` datasets ranked by JGD-score. We highlight the best model in **bold** and underline the second best.

Dataset	JGD	GRU-ODE	LinODEnet	CRU	Neural Flows	GraFITi	GraFITi-C
DUP01	2.697	1.037± 0.047	0.964± 0.036	0.958± 0.038	1.030± 0.046	0.955± 0.037	**0.951± 0.036**
JEL01	2.580	1.017± 0.014	0.949± 0.015	0.939± 0.015	1.000± 0.013	0.942± 0.020	**0.935± 0.016**
DOK01	2.277	1.011± 0.004	0.996± 0.003	0.985± 0.005	0.998± 0.005	0.984± 0.005	**0.982± 0.005**
INA01	2.218	1.018± 0.010	1.009± 0.011	1.005± 0.010	1.008± 0.008	**1.004± 0.010**	**1.004± 0.009**
WOL01	1.973	0.952± 0.035	0.806± 0.027	0.814± 0.029	0.841± 0.029	0.787± 0.028	**0.784± 0.030**
BOR01	1.795	0.794± 0.034	0.719± 0.021	0.715± 0.020	0.743± 0.026	0.712± 0.022	**0.709± 0.022**
HYN01	1.548	0.883± 0.060	0.672± 0.044	0.665± 0.053	0.636± 0.045	0.625± 0.043	**0.619± 0.046**
JEL02	1.271	0.816± 0.049	0.693± 0.031	**0.674± 0.028**	0.779± 0.028	0.699± 0.029	0.687± 0.027
DUP02	1.202	0.895± 0.055	0.740± 0.042	0.722± 0.046	0.890± 0.081	0.728± 0.044	**0.718± 0.046**
WOL02	0.895	0.854± 0.010	0.663± 0.015	0.653± 0.017	0.685± 0.012	0.654± 0.014	**0.645± 0.016**
DIF01	0.735	1.035± 0.023	**0.832± 0.087**	0.985± 0.025	1.014± 0.025	0.985± 0.030	0.982± 0.029
VAN01	0.407	0.321± 0.023	0.250± 0.006	0.253± 0.005	0.250± 0.006	0.246± 0.005	**0.242± 0.006**
DUP03	0.254	0.874± 0.089	0.632± 0.044	**0.622± 0.047**	1.098± 0.447	0.627± 0.043	0.744± 0.042
BER01	0.179	0.594± 0.054	**0.279± 0.020**	0.280± 0.016	0.398± 0.014	0.300± 0.018	0.342± 0.018
LEN01	0.178	1.028± 0.060	**0.387± 0.071**	0.754± 0.157	1.009± 0.061	0.607± 0.055	0.970± 0.063
LI01	0.113	1.067± 0.015	**0.084± 0.009**	0.175± 0.020	0.979± 0.049	0.202± 0.013	0.742± 0.010
LI02	0.097	0.723± 0.080	0.434± 0.044	0.437± 0.046	0.674± 0.105	**0.397± 0.058**	0.458± 0.056
REV01	0.081	1.091± 0.075	**0.597± 0.061**	0.602± 0.049	0.978± 0.042	0.674± 0.055	0.855± 0.050
PUR01	0.049	0.703± 0.058	**0.106± 0.006**	0.353± 0.083	0.654± 0.102	0.153± 0.006	0.476± 0.020
NYG01	0.047	0.571± 0.074	0.358± 0.071	0.403± 0.092	0.442± 0.094	**0.344± 0.065**	0.366± 0.047
PUR02	0.044	0.830± 0.066	**0.280± 0.028**	0.293± 0.026	0.723± 0.077	0.322± 0.021	0.511± 0.023
HOD01	0.042	0.851± 0.118	0.441± 0.043	**0.409± 0.049**	0.701± 0.077	0.493± 0.046	0.609± 0.056
REE01	0.035	0.266± 0.068	0.045± 0.012	0.051± 0.008	0.045± 0.011	**0.033± 0.007**	0.039± 0.012
VIL01	0.028	0.511± 0.053	0.374± 0.021	0.373± 0.039	0.500± 0.060	**0.344± 0.044**	0.378± 0.042
KAR01	0.023	0.193± 0.013	**0.034± 0.008**	0.044± 0.012	0.069± 0.009	0.041± 0.013	0.078± 0.011
SHO01	0.023	0.260± 0.017	0.057± 0.006	0.095± 0.010	0.092± 0.014	0.062± 0.013	**0.055± 0.013**
BUT01	0.020	0.583± 0.172	**0.254± 0.074**	0.317± 0.108	0.441± 0.153	0.281± 0.071	0.324± 0.091
MAL01	0.019	0.420± 0.061	**0.018± 0.007**	0.064± 0.007	0.052± 0.002	0.020± 0.004	0.054± 0.005
ASL01	0.019	0.114± 0.035	**0.022± 0.003**	0.046± 0.014	0.066± 0.033	0.025± 0.009	0.026± 0.002
BUT02	0.016	0.483± 0.047	**0.207± 0.056**	0.282± 0.042	0.329± 0.068	0.248± 0.052	0.256± 0.039
MIT01	0.015	0.005± 0.001	**0.003± 0.000**	**0.003± 0.000**	0.008± 0.004	**0.003± 0.000**	**0.003± 0.000**
GUP01	0.014	0.153± 0.041	**0.018± 0.007**	0.057± 0.017	0.043± 0.017	0.041± 0.006	0.035± 0.006
GUY01	0.013	0.036± 0.004	0.006± 0.005	**0.004± 0.001**	0.024± 0.015	0.005± 0.003	**0.004± 0.001**
PHI01	0.013	0.674± 0.030	**0.131± 0.014**	0.133± 0.020	0.635± 0.158	0.222± 0.013	0.345± 0.015
GUY02	0.013	0.124± 0.044	**0.010± 0.006**	**0.010± 0.002**	0.082± 0.066	0.012± 0.009	0.032± 0.015
PUL01	0.013	0.099± 0.024	**0.008± 0.004**	0.012± 0.003	0.061± 0.060	**0.008± 0.001**	0.024± 0.008
CAL01	0.013	1.049± 0.055	**0.078± 0.009**	0.158± 0.008	0.867± 0.014	0.179± 0.012	0.643± 0.024
WOD01	0.012	0.612± 0.072	0.154± 0.016	**0.113± 0.017**	0.510± 0.060	0.164± 0.013	0.344± 0.016
GUP02	0.012	0.870± 0.059	0.469± 0.022	**0.444± 0.018**	0.577± 0.017	0.449± 0.027	0.461± 0.025
M01	0.012	0.055± 0.009	0.004± 0.001	0.005± 0.000	0.124± 0.207	**0.003± 0.000**	**0.003± 0.000**
LEN02	0.012	0.297± 0.071	**0.039± 0.005**	0.059± 0.012	0.380± 0.141	0.099± 0.021	0.143± 0.022
KAR02	0.011	0.515± 0.032	**0.140± 0.010**	0.151± 0.011	0.257± 0.016	0.151± 0.009	0.252± 0.010
SHO02	0.011	0.368± 0.028	**0.037± 0.006**	0.083± 0.015	0.109± 0.015	0.043± 0.006	0.073± 0.010
MAC01	0.010	0.242± 0.026	0.020± 0.003	0.065± 0.006	0.029± 0.011	0.021± 0.003	**0.019± 0.002**
IRI01	0.010	0.151± 0.032	**0.037± 0.003**	0.049± 0.010	0.116± 0.006	0.038± 0.017	0.097± 0.008
BAG01	0.010	0.294± 0.041	0.032± 0.005	0.046± 0.005	0.075± 0.012	**0.029± 0.002**	0.109± 0.002
WOL03	0.008	0.859± 0.114	**0.073± 0.010**	0.177± 0.016	0.479± 0.107	0.105± 0.016	0.247± 0.032
WAN01	0.008	0.504± 0.031	**0.103± 0.012**	0.125± 0.012	0.345± 0.027	0.119± 0.010	0.232± 0.015
NEL01	0.007	0.054± 0.012	0.010± 0.001	0.009± 0.001	0.060± 0.029	**0.007± 0.000**	0.023± 0.006
HUA01	0.007	0.321± 0.046	**0.052± 0.004**	0.116± 0.007	0.149± 0.015	0.063± 0.005	0.115± 0.007
# Wins	—	0	**25**	8	2	10	15
Rank	—	5.9	**2.10**	2.82	4.82	2.20	2.86

Consequently, we build the benchmark on top of real-world ODEs: their system, their constants and their initial values establish a real-world connection that has been created in hundreds of scientific publications. To delineate our benchmark from purely synthetic data we therefore call it *semi-synthetic*.

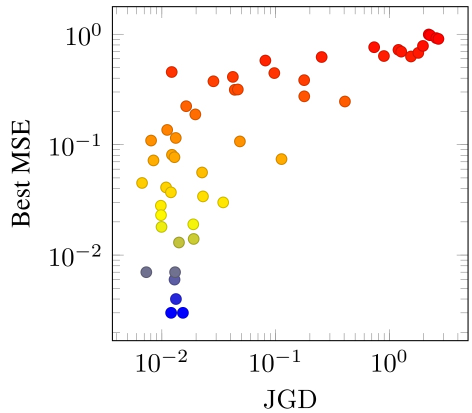

Figure 2: Test MSE of the best performing model vs JGD-score across 50 datasets.

The fact that the datasets are generated by ODEs implies that some general ODE-based models are expressive enough to mimic the generating process and to perform well for large amounts of data. For example, LinODEnet can represent a Koopman-type (Koopman, 1931; Koopman and v. Neumann, 1932) linear approximation of any nonlinear ODE under some assumptions. However, being expressive enough does not mean that these models are best when trained on a limited amount of data. Other, non-ODE models can beat ODE-based models with limited training data, even if the ground truth is an ODE. In fact, a main purpose of the benchmark is to answer questions about the best forecasting models for data generated from an (unknown) ODE ground truth. Despite our limitation to ODEs from biology the perspective is forecasting for many scientific domains where dynamics are *in principle* governed by ODEs.

The inductive bias of ODE-based models (LinODEnet (Scholz et al., 2022)) is rather weak. Being based on ODEs does not mean that these models would only perform well on data generated from ODEs. When one spells out the expressiveness of LinODEnet in terms of basis functions, it would just mean that after a nonlinear transformation the time dependent functions can be expressed by some basis functions given as solutions of some linear ODE. The representation is entirely trainable and nothing about the time series in question is hard-coded apart from smoothness.

The number of instances per dataset contained in `Physiome-ODE` is deliberately small (2000). Our goal was not to create huge datasets just for the sake of being huge, but to create datasets for the purpose of research: the smallest datasets that would allow complex, ODE-based models to demonstrate their strengths. However, it is trivial to create an arbitrary number of time series, if larger datasets are desired.

While this work focuses on IMTS forecasting, `Physiome-ODE` can be easily extended to regular time series forecasting. In Appendix G, we conduct experiments with three datasets created with `Physiome-ODE`. Our results show that these datasets have an interesting property: They force models to learn channel dependencies. This contradicts with the results on traditional evaluation datasets, in which models like PatchTST (Nie et al., 2023) are more effective when they ignore inter-channel correlations.

7 Prior Time Series Forecasting Benchmarks

Monash time series forecasting archive (Godahewa et al., 2021) The Monash repository is the first benchmark for time series forecasting. It encompasses 11 openly-accessible multivariate time series datasets drawn from various domains, including energy, banking, and nature. Some of these datasets may include missing values. However, the sampling process is regular. Therefore, the missing values can easily be imputed, so that methods designed for regular time series forecasting can be applied. Hypothetically, we could use the 11 multivariate time series contained in the Monash archive to create IMTS datasets for an IMTS forecasting benchmark. However, most of these multivariate datasets are actually multiple univariate time series stacked on top of each other and/or a single long time series that is segmented to create multiple time series instances. Other datasets used to evaluate regular multivariate time series forecasting (Nie et al., 2023; Zeng et al., 2023; Zhou et al., 2021), are not well-suited for IMTS forecasting for similar reasons. In fact, Nie et al. (2023) showed that models benefit, when they ignore channel correlations, hinting that these are barely carrying useful information in the respective datasets. In IMTS forecasting however, modeling these channel correlations is a crucial property of the models, due to the high number of missing values.

Chaotic ODEs benchmark for time series forecasting Closely related to `Physiome-ODE`, Gilpin (2021) provides a forecasting benchmark created with chaotic ordinary differential equations that mainly originate from Physics. Similar to the Monash repository, each chaotic ODE creates single long trajectory, without varying the constants. In contrast, `Physiome-ODE` contains 2000 different trajectories with varying constants and initial states. Furthermore, the inherent chaotic nature of these equations often presents a forecasting challenge, even in the regularly sampled and fully observed forecasting setting. On the other hand, IMTS forecasting is challenging due to the sparse observation of a time series. We assume that combining these two factors of difficulty, would surpass any current methods forecast modeling ability. This is closely related to what we observe on the most difficult dataset of `Physiome-ODE`. We support our assumption, by an example experiment on the Lorenz-attractor, a famous chaotic ODE, which is a part of Gilpin (2021)'s benchmark. Respective experimental details are given in Appendix H and results in Table 7. Similarly, to certain `Physiome-ODE` datasets with a high JGD (e.g. DUP01), no method can significantly outperform the simple baseline GraFITi-C.

PDEBench (Takamoto et al., 2022) Related to applying differential equations to create datasets, there exists a benchmark with 11 spatio-temporal datasets named PDEBench. Here, models are trained to estimate the parameters of an underlying partial differential equation (PDE) used to generate such dataset, instead of forecasting. Current IMTS forecasting architectures cannot utilize the spatio-temporal information inherent in PDEBench's datasets.

8 Conclusion

We introduced `Physiome-ODE`, the first ever IMTS forecasting benchmark. `Physiome-ODE`'s datasets are created with ODE models, that were defined in decades of research and published on the Physiome Model Repository. Our experiments showed that LinODEnet and CRU are actually better than previous evaluation on established datasets indicated. Nevertheless, we also provided a few datasets, on which models are unable to outperform a constant baseline model. We believe that our datasets, especially the very difficult ones, can help to identify deficits of current architectures and support future research on IMTS forecasting.

References

D. W. K. Andrews. Consistency in Nonlinear Econometric Models: A Generic Uniform Law of Large Numbers. *Econometrica*, 55(6):1465, Nov. 1987. ISSN 00129682. doi: 10.2307/1913568. URL `https://www.jstor.org/stable/1913568?origin=crossref`.

S.-A. Chen, C.-L. Li, N. Yoder, S. O. Arik, and T. Pfister. TSMixer: An All-MLP Architecture for Time Series Forecasting, Sept. 2023.

R. I. Jennrich. Asymptotic Properties of Non-Linear Least Squares Estimators. *The Annals of Mathematical Statistics*, 40(2):633–643, 1969. ISSN 0003-4851. doi: 10.1214/aoms/1177697731.

A. Johnson, L. Bulgarelli, T. Pollard, S. Horng, L. A. Celi, and R. Mark. MIMIC-IV. *PhysioNet*, 2021. doi: 10.13026/RRGF-XW32.

A. E. W. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, and R. G. Mark. MIMIC-III, a freely accessible critical care database. *Scientific Data*, 3(1):160035, May 2016. ISSN 2052-4463. doi: 10.1038/sdata.2016.35.

R. E. Kalman. A New Approach to Linear Filtering and Prediction Problems. *Journal of Basic Engineering*, 82(1):35–45, Mar. 1960. ISSN 0021-9223. doi: 10.1115/1.3662552.

M. J. Menne, J. Williams, and R. S. Vose. Long-Term Daily and Monthly Climate Records from Stations Across the Contiguous United States (U.S. Historical Climatology Network). Technical Report osti:1394920; cdiac:NDP-019; doi:10.3334/CDIAC/CLI.NDP019, Environmental System Science Data Infrastructure for a Virtual Ecosystem (ESS-DIVE) (United States); CDIAC, Jan. 2016.

F. M. Mione, L. Kaspersetz, M. F. Luna, J. Aizpuru, R. Scholz, M. Borisyak, A. Kemmer, M. T. Schermeyer, E. C. Martinez, P. Neubauer, and M. N. Cruz Bournazou. A workflow management system for reproducible and interoperable high-throughput self-driving experiments. *Computers & Chemical Engineering*, 187:108720, Aug. 2024. ISSN 0098-1354. doi: 10.1016/j.compchemeng.2024.108720.

R. Scholz, S. Born, N. Duong-Trung, M. N. Cruz-Bournazou, and L. Schmidt-Thieme. Latent Linear ODEs with Neural Kalman Filtering for Irregular Time Series Forecasting. Sept. 2022.

I. Silva, G. Moody, D. J. Scott, L. A. Celi, and R. G. Mark. Predicting in-hospital mortality of ICU patients: The PhysioNet/Computing in cardiology challenge 2012. In *2012 Computing in Cardiology*, pages 245–248, Sept. 2012.

M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. Pdebench: An extensive benchmark for scientific machine learning. *Advances in Neural Information Processing Systems*, 35:1596–1611, 2022.

G. Teschl. *Ordinary Differential Equations and Dynamical Systems*. American Mathematical Soc., Aug. 2012. ISBN 978-0-8218-8328-0.

T. Yu, C. M. Lloyd, D. P. Nickerson, M. T. Cooling, A. K. Miller, A. Garny, J. R. Terkildsen, J. Lawson, R. D. Britten, P. J. Hunter, and P. M. F. Nielsen. The Physiome Model Repository 2. *Bioinformatics*, 27(5):743–744, Mar. 2011. ISSN 1367-4803. doi: 10.1093/bioinformatics/btq723.

A. Zeng, M. Chen, L. Zhang, and Q. Xu. Are Transformers Effective for Time Series Forecasting? In *AAAI*, 2023.

A Real World Dataset Statistics

Table 3: Statistics of evaluation datasets used by De Brouwer et al. (2019); Biloš et al. (2021); Schirmer et al. (2022). *Max. Len.* refers to the maximum sequence length among samples. *Max. Obs.* refers to the maximum number of non-missing observations among samples. *Sparsity* refers to the percentage of missing values over all samples.

name	Instances	Channel	Max. Len	Max. Obs	Spars.
USHCN	1.114	5	370	398	78.0%
PhysioNet-2012	11.981	37	48	606	80.4%
MIMIC-III	21.250	96	97	677	94.2%
MIMIC-IV	17.874	102	920	1642	97.8%

B Proofs

B.1 Proof of Lemma 1

Proof. Due to Equation (5), it is sufficient to show that

$$\widehat{\mathrm{std}}[x^{\mathrm{diff},\epsilon}]^2 \xrightarrow{\epsilon\to 0} \frac{1}{T}\int_{t=0}^{T}\left(x'(t) - \frac{(x(T)-x(0))}{T}\right)^2 \mathrm{d}t. \tag{13}$$

Due to the mean value theorem, there exists $\zeta_{t,\epsilon} \in [t-\epsilon, t]$ with $\frac{x(t)-x(t-\epsilon)}{\epsilon} = x'(\zeta_{t,\epsilon})$ for all $t = \epsilon, 2\epsilon, \ldots, T$. Given $N \coloneqq \frac{T}{\epsilon}$ and $\mu_N \coloneqq \frac{1}{N}\sum_{t=\epsilon,2\epsilon,\ldots,T} x_t^{\mathrm{diff},\epsilon}$, we thus have

$$\begin{aligned}
\mu_N &= \frac{1}{N}\sum_{t=\epsilon,2\epsilon,\ldots,T} x'(\zeta_{t,\epsilon}) && \text{(14a)}\\
&= \frac{1}{T}\sum_{t=\epsilon,2\epsilon,\ldots,T} \underbrace{\frac{T}{N}}_{=\epsilon} x'(\zeta_{t,\epsilon}) && \text{(14b)}\\
&\xrightarrow{\epsilon\to 0} \frac{1}{T}\int_{t=0}^{T} x'(t)\,\mathrm{d}t && \text{(14c)}\\
&= \frac{x(T)-x(0)}{T} && \text{(14d)}
\end{aligned}$$

due to the definition of the Riemann integral. Furthermore, we have

$$\widehat{\mathrm{std}}[x^{\mathrm{diff},\epsilon}]^2 = \frac{1}{N}\Big(\sum_{t=\epsilon,2\epsilon,\ldots,T} (x_t^{\mathrm{diff},\epsilon})^2\Big) - \mu_N^2 \tag{15}$$

Therefore, again by the mean value theorem and the definition of the Riemann integral:

$$\begin{aligned}
\frac{1}{N}\Big(\sum_{t=\epsilon,2\epsilon,\ldots,T} (x_t^{\mathrm{diff},\epsilon})^2\Big) &= \frac{1}{N}\Big(\sum_{t=\epsilon,2\epsilon,\ldots,T} x'(\zeta_{t,\epsilon})^2\Big) && \text{(16a)}\\
&= \frac{1}{T}\sum_{t=\epsilon,2\epsilon,\ldots,T} \underbrace{\frac{T}{N}}_{=\epsilon} x'(\zeta_{t,\epsilon})^2 && \text{(16b)}\\
&\xrightarrow{\epsilon\to 0} \frac{1}{T}\int_{t=0}^{T} x'(t)^2\,\mathrm{d}t && \text{(16c)}
\end{aligned}$$

Consequently,

$$\begin{aligned}
\widehat{\mathrm{std}}[x^{\mathrm{diff},\epsilon}]^2 &\xrightarrow{\epsilon\to 0} \frac{1}{T}\int_{t=0}^{T} x'(t)^2\,\mathrm{d}t - \left(\frac{(x(T)-x(0))}{T}\right)^2 && \text{(17)}\\
&= \frac{1}{T}\int_{t=0}^{T}\left(x'(t) - \frac{(x(T)-x(0))}{T}\right)^2 \mathrm{d}t. && \text{(18)}
\end{aligned}$$

□

B.2 Assumptions and Proof of Lemma 2

For the lemma to hold we make the technical assumption:

(A) The distribution p of continuous functions $[0,T] \to \mathbb{R}$ is given by a continuous map[2] $\phi\colon \Theta \times [-1,T] \to \mathbb{R}, (\theta,t) \mapsto \phi(\theta,t)$, with a compact parameter space $\Theta \subset \mathbb{R}^d$, and a probability measure μ on Θ. We require that the map $\frac{\partial\phi}{\partial t}$ is continuous, too.

We sometimes use x_θ for $\phi(\theta,\cdot)$ and x'_θ for $\frac{\partial\phi}{\partial t}(\theta,\cdot)$. This is no restriction for the differential equations we consider, as we can always obtain a solution on $[-1,T]$, given parameters (including $x(0)=x_0$).

Sampling from this distribution is done by sampling from the probability measure on Θ. A sequence of i.i.d. random variables θ_n, $n \in \mathbb{N}_{>0}$, distributed according to μ, defines a sequence of random functions $(x_n)_{n\in\mathbb{N}_{>0}}$, $x_n(t) = \phi(\theta_n,t)$, as in the statement of the Lemma 2.

For the proof we will need a uniform law of large numbers (ULLN). There exist many versions of such theorems (e.g. (Jennrich, 1969), or (Andrews, 1987)), but we prefer to state and prove a version with strong assumptions appropriate for our setting.

Lemma 3 (Uniform Law of Large Numbers)**.** *Let $f\colon \Theta \times I \to \mathbb{R}$ be a continuous function on compact metric spaces Θ and I. Let θ_i, $i \in \mathbb{N}_{>0}$ be i.i.d. random variables with values in Θ, all distributed according to a probability measure μ on Θ. Then*

$$\frac{1}{N}\sum_{i=1}^{N} f(\theta_i,t) \xrightarrow{N\to\infty} \int_\Theta f(\theta,t)\,\mathrm{d}\mu(\theta) \tag{19}$$

uniformly in $t \in I$ almost surely.

Proof. As $\Theta \times I$ is compact and f continuous, f is even uniformly continuous. Hence, for any $\epsilon > 0$, we can then choose a δ such that

$$d(\theta,\theta') < \delta \quad \text{and} \quad (t,t') < \delta \Longrightarrow |f(\theta',t') - f(\theta,t)| < \epsilon \tag{20}$$

Where d is the metric on Θ. Now by compactness of I, finitely many δ-balls in I cover it, i.e. there are $t_1,\ldots,t_k \in I$, such that the $B_\delta(t_j)$, $j=1,\ldots,k$ cover I. For each $(t_j)_{j=1:k}$ we can apply a strong law of large numbers to obtain almost surely convergence of $\frac{1}{N}\sum_{i=1}^{N} f(\theta_i,t_j)$ to $\int_\Theta f(\theta,t_j)\,\mathrm{d}\mu(\theta)$.

Hence, almost surely, the following holds: For $\epsilon > 0$ we can choose $N_0 > 0$ such that for all natural numbers $N \geq N_0$ and all $j \in \{1,\ldots,k\}$:

$$\left|\frac{1}{N}\sum_{i=1}^{N} f(\theta_i,t_j) - \int_\Theta f(\theta,t_j)\,\mathrm{d}\mu(\theta)\right| < \epsilon \tag{21}$$

Now for any $t \in I$ there is a $j \in \{1,\ldots,k\}$ such that

$$\left|\frac{1}{N}\sum_{i=1}^{N}\big(f(\theta_i,t_j) - f(\theta_i,t)\big)\right| < \epsilon \qquad \text{and} \qquad \left|\int_\Theta f(\theta,t_j) - f(\theta,t)\,\mathrm{d}\mu(\theta)\right| < \epsilon \tag{22}$$

The triangle inequality yields

$$\left|\frac{1}{N}\sum_{i=1}^{N} f(\theta_i,t) - \int_\Theta f(\theta,t)\,\mathrm{d}\mu(\theta)\right| < 3\epsilon \tag{23}$$

for all $t \in I$, which proves the claim. $\square$

Proof. (Lemma 2) Let θ_i, $i \in \mathbb{N}_{>0}$ be i.i.d. random variables with values in Θ, all distributed according to a probability measure μ on Θ.

[2] We define ϕ on the larger interval $[-1,T]$, instead of $[0,T]$, in order to use one-sided difference quotients.

Define $\psi : \Theta \times [0,1] \times [0,T]$ by

$$\psi(\theta, \epsilon, t) = \begin{cases} \frac{\phi(\theta,t)-\phi(\theta,t-\epsilon)}{\epsilon} & \text{if } \epsilon > 0 \\ \frac{\partial \phi}{\partial t}(\theta, t) & \text{if } \epsilon = 0 \end{cases} \tag{24}$$

For $\epsilon > 0$ the function is obviously continuous as a composition of continuous functions. Let now be $\epsilon = 0$ and $t \in [0,T]$, $\theta \in \Theta$. We claim that ψ is continuous at $(\theta, 0, t)$. $\frac{\partial \phi}{\partial t}$ is uniformly continuous on its compact domain of definition.

Let δ be greater than zero. We can choose a $\gamma > 0$ such that for $d(\theta, \theta') < \gamma$ and $|t - t'| < \gamma$ we have $\left| \frac{\partial \phi}{\partial t}(\theta', t') - \frac{\partial \phi}{\partial t}(\theta, t) \right| < \delta$.

Now chose (θ', ϵ', t') with $d(\theta, \theta') < \gamma$ and $d(t, t') < \gamma/2$ and $0 \leq \epsilon' < \gamma/2$. If $\epsilon' = 0$, we directly obtain

$$|\psi(\theta', 0, t') - \psi(\theta, 0, t)| = \left| \frac{\partial \phi}{\partial t}(\theta', t') - \frac{\partial \phi}{\partial t}(\theta, t) \right| < \delta \tag{25}$$

If $\epsilon' > 0$, the mean value theorem implied that there is a $\tau \in (t' - \epsilon', t')$ such that:

$$|\psi(\theta', \epsilon', t') - \psi(\theta, 0, t)| = \left| \frac{\partial \phi}{\partial t}(\theta', \tau) - \frac{\partial \phi}{\partial t}(\theta, t) \right| < \delta \tag{26}$$

as $d(\tau, t) \leq d(\tau, t') + d(t', t) \leq \epsilon' + \gamma/2 < \gamma/2 + \gamma/2 = \gamma$. This establishes the continuity of ψ.

Therefore, ψ is also uniformly continuous on its compact domain of definition.

Define $\Psi \colon [0,1] \times [0,T] \to \mathbb{R}$ by

$$\Psi(\epsilon, t) = \int_\Theta \psi(\theta, \epsilon, t) \, \mathrm{d}\mu(\theta) \tag{27}$$

We also define $\Psi^N \colon [0,1] \times [0,T] \to \mathbb{R}$

$$\Psi^N(\epsilon, t) = \frac{1}{N} \sum_{i=1}^{N} \psi(\theta_i, \epsilon, t) \tag{28}$$

The uniform law of large numbers shows that almost surely Ψ^N converges uniformly to Ψ for $N \to \infty$.

Analogously we define $\Xi \colon [0,1] \to \mathbb{R}$ by

$$\Xi(\epsilon, t) = \int_\Theta \psi(\theta, \epsilon, t)^2 \, \mathrm{d}\mu(\theta) \tag{29}$$

We also define $\Psi^N \colon [0,1] \times [0,T] \to \mathbb{R}$

$$\Xi^N(\epsilon, t) = \frac{1}{N} \sum_{i=1}^{N} \psi(\theta_i, \epsilon, t)^2 \tag{30}$$

Again, the uniform law of large numbers shows that almost surely Ξ^N converges uniformly to Ξ for $N \to \infty$. Hence,

$$\widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}] = \sqrt{\Xi^N(\epsilon, t) - \Psi^N(\epsilon, t)^2} \tag{31}$$

converges almost surely uniformly to $\sqrt{\Xi(\epsilon, t) - \Psi(\epsilon, t)^2} = \mathrm{std}_{\theta \sim \mu}[\Psi(\epsilon, t)]$ for $N \to \infty$.

On the other hand by uniform continuity

$$\widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}] \xrightarrow[\text{uniformly}]{\epsilon \to 0} \widehat{\mathrm{std}}[(\psi(\theta_i, 0, t))_{i=(1:N)}] \tag{32}$$

$$\sqrt{\Xi(\epsilon, t) - \Psi(\epsilon, t)^2} \xrightarrow[\text{uniformly}]{\epsilon \to 0} \mathrm{std}_{\theta \sim \mu}[(\Psi(0, t))] = \mathrm{std}_{x \sim p} \, x'(t) \tag{33}$$

Now, given some $\delta > 0$, we can choose an ϵ' and almost surely an N_0 such that for all $0 < \epsilon < \epsilon'$ and $N \geq N_0$:

$$\left|\widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}] - \widehat{\mathrm{std}}[(\psi(\theta_i, 0, t))_{i=(1:N)}]\right|_{C^0([0,T])} \leq \delta \tag{34}$$

$$\left|\widehat{\mathrm{std}}[(\psi(\theta_i, 0, t))_{i=(1:N)}] - \mathrm{std}_{x\sim p}\, x'(t)\right|_{C^0([0,T])} \leq \delta \tag{35}$$

and for $t, t' \in [0, T]$, $|t - t'| < \epsilon$:

$$\left|\widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}] - \widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t'))_{i=(1:N)}]\right| \leq \delta \tag{36}$$

Therefore,

$$\left|\frac{1}{T}\int_0^T \widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}]\, \mathrm{d}t - \frac{1}{T}\int_0^T \mathrm{std}_{x\sim p}\, x'(t)\, \mathrm{d}t\right| < 2\delta \tag{37}$$

and for $T/\epsilon \in \mathbb{N}$ the Riemann sum with step size ϵ differs from the integral by at most $T\delta$, i.e.

$$\left|\frac{1}{T}\int_0^T \widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, t))_{i=(1:N)}]\, \mathrm{d}t - \frac{\epsilon}{T}\sum_{k=1}^{T/\epsilon} \widehat{\mathrm{std}}[(\psi(\theta_i, \epsilon, k\epsilon))_{i=(1:N)}]\right| < \delta \tag{38}$$

In the notations of the lemma, with $\frac{1}{T}\int_0^T \mathrm{std}_{x\sim p}\, x'(t)\, \mathrm{d}t = \mathrm{MPGD}(p)$ we get:

$$\left|\frac{\epsilon}{T}\sum_{t=\epsilon}^{T} \widehat{\mathrm{std}}[(x_{i,t}^{\mathrm{diff}})_{i=1:N}] - \mathrm{MPGD}(p)\right| < 3\delta \tag{39}$$

Thus proving the claim that the left term of the difference converges almost surely for $\epsilon \to 0$ and $N \to \infty$ to the right term of the difference. Please note that N and ϵ were chosen independently. The limits can be taken simultaneously or in arbitrary order. $\square$

C Relation of MGD and Lipschitz Constant

Regarding the relationship between the MGD and the Lipschitz constant of the vector field, we can consider the standard example of a linear ODE: $\dot{x}(t) = a \cdot x(t)$, for which the Lipschitz constant is precisely a. For simplicity, assume the initial condition $x(0) = 1$. Hence, the solution is $x(t) = e^{at}$. Then, by (4) we have $c = \frac{1}{T}(e^{aT} - 1)$, and hence:

$$\begin{aligned}
\frac{1}{T}\int_0^T \|\dot{x}(t) - c\|^2 \, dt &= \left(\frac{a}{2T} - \frac{1}{T^2}\right) e^{2aT} + \frac{1}{2T^2}e^{aT} - \frac{a}{2T} - \frac{1}{T^2} \\
&= \mathcal{O}\left(\frac{a}{T}e^{2aT}\right) \\
\implies \text{MGD} &= \mathcal{O}\left(\sqrt{\frac{a}{T}}e^{aT}\right)
\end{aligned}$$

In particular, for this example, the MGD grows super-exponentially with the Lipschitz constant. This result can be extended to an upper bound for general systems:

1. By the Mean Value Theorem, there exists $\xi \in (0, T)$ such that $c = \frac{x(T)-x(0)}{T} = \dot{x}(\xi) = f(x(\xi))$.

2. Therefore, $\|\dot{x}(t) - c\|^2 = \|\dot{x}(t) - \dot{x}(\xi)\|^2 = \|f(x(t)) - f(x(\xi))\|^2 \leq L^2\|x(t) - x(\xi)\|^2$.

3. We can bound the latter quantity via Grönwall's Lemma:

$$\begin{aligned}
\|x(t) - x(\xi)\| &= \left\|\int_\xi^t \dot{x}(s)\, ds\right\| \\
&= \left\|\int_\xi^t f(x(s))\, ds\right\| \\
&\leq \int_\xi^t \|f(x(s))\|\, ds \quad \text{assuming } t \geq \xi \\
&\leq \int_\xi^t \|f(x(s)) - f(x(\xi))\| + \|f(x(\xi))\|\, ds \\
&\leq \underbrace{(t-\xi) \cdot \|f(x(\xi))\|}_{=\alpha(t)} + \int_\xi^t \underbrace{L}_{=\beta(s)} \cdot \underbrace{\|x(s) - x(\xi)\|}_{=u(s)}\, ds \\
&\leq (t-\xi) \cdot \|f(x(\xi))\| \cdot e^{L(t-\xi)} \quad \text{via Grönwall}
\end{aligned}$$

And if $\xi \geq t$, we analogously get $\|x(t) - x(\xi)\| \leq (\xi - t) \cdot \|f(x(\xi))\| \cdot e^{L(\xi - t)}$.

In particular, note that by the same method:

$$\|f(x(\xi))\| = \frac{1}{T}\|x(T) - x(0)\| \leq \|f(x(0))\|e^{LT}$$

5. We use this result to bound the integral:

$$\begin{aligned}
\frac{1}{T}\int_0^T \|\dot{x}(t)-c\|^2\,dt &\leq \frac{L^2}{T}\int_0^T \|x(t)-x(\xi)\|^2\,dt \\
&= \frac{L^2}{T}\left(\int_0^\xi \|x(t)-x(\xi)\|^2\,dt + \int_\xi^T \|x(t)-x(\xi)\|^2\,dt\right) \\
&\leq \frac{L^2}{T}\|f(x(\xi))\|^2 \cdot \left(\int_0^\xi (\xi-t)^2 \cdot e^{2L(\xi-t)}\,dt + \int_\xi^T (t-\xi)^2 \cdot e^{2L(t-\xi)}\,dt\right) \\
&\leq \ldots \\
&\leq \|f(x(\xi))\|^2\left((2LT^2 - T + \frac{1}{2L})e^{2LT} - \frac{1}{2L}\right) \\
&\lesssim \|f(x(0))\|^2 e^{2LT}\mathcal{O}(LT^2e^{2LT}) \\
&= \mathcal{O}\left(\|f(x(0))\|^2 LTe^{4LT}\right)
\end{aligned}$$

6. Plugging this back into the definition of the MGD, we get:

$$\text{MGD} = \sqrt{\frac{1}{T}\int_0^T \|\dot{x}(t)-c\|^2\,dt} = \mathcal{O}(\sqrt{LT}e^{2LT})$$

D Dataset Description

Table 4: Optimized Datasets created with generated Python code published on Physiome. A t-unit of "—" refers to Physiome giving the time unit as dimensionless. We entered — in σ_{dur}, σ_{state}, σ_{const} and JGD, if we were unable to create a dataset with our method due to errors thrown by the ode solver or the occurence of ode-explosions (see Section 4)

Model	Domain	Channel	Constants	Time Unit	T	d-state	d-const	ATVL
DUP01	Calcium-Dynamics	2	13	s	30.0	0.3	0.05	2.697
JEL01	Endocrine	2	15	-	30.0	0.5	0.05	2.58
DOK01	Electrophysiology	18	46	s	10.0	0.1	0.05	2.277
INA01	Electrophysiology	29	58	s	10.0	0.5	0.05	2.218
WOL01	Signal-Transduction	9	19	min	10.0	0.3	0.05	1.973
BOR01	Calcium-Dynamics	3	14	min	30.0	0.1	0.05	1.795
HYN01	Metabolism	22	60	min	30.0	0.1	0.05	1.548
JEL02	Endocrine	2	15	-	30.0	0.5	0.05	1.271
DUP02	Calcium-Dynamics	3	19	min	30.0	0.1	0.05	1.202
WOL02	Metabolism	7	14	min	30.0	0.1	0.1	0.895
DIF01	Electrophysiology	16	50	s	10.0	0.1	0.1	0.735
VAN01	Metabolism	10	38	s	10.0	0.1	0.05	0.407
DUP03	Calcium-Dynamics	2	14	s	10.0	0.3	0.05	0.254
BER01	Signal-Transduction	11	22	ms	30.0	0.1	0.3	0.179
LEN01	Endocrine	3	7	s	30.0	0.1	0.1	0.178
LI01	Electrophysiology	5	20	s	30.0	0.3	0.05	0.113
LI02	Electrophysiology	5	20	s	10.0	0.1	0.05	0.097
REV01	Immunology	5	10	day	30.0	0.5	0.3	0.081
PUR01	Neurobiology	3	21	ms	30.0	0.1	0.05	0.049
NYG01	Electrophysiology	29	51	s	0.33	0.5	0.1	0.047
PUR02	Neurobiology	3	21	ms	30.0	0.1	0.05	0.044
HOD01	Electrophysiology	4	8	ms	30.0	0.5	0.3	0.042
REE01	Metabolism	4	21	h	30.0	0.5	0.05	0.035
VIL01	Signal-Transduction	9	16	h	30.0	0.1	0.3	0.028
KAR01	Signal-Transduction	16	21	s	10.0	0.3	0.05	0.023
SHO01	Excitation-contraction-Coupling	56	105	ms	30.0	0.3	0.05	0.023
BUT01	Electrophysiology	4	24	ms	30.0	0.3	0.05	0.02
MAL01	Endocrine	9	51	day	30.0	0.5	0.05	0.019
ASL01	Electrophysiology	30	83	ms	30.0	0.3	0.3	0.019
BUT02	Electrophysiology	4	25	ms	30.0	0.3	0.05	0.016
MIT01	Immunology	7	13	day	30.0	0.1	0.3	0.015
GUP01	Endocrine	4	7	h	30.0	0.5	0.05	0.014
GUY01	Cardiovascular-Circulation	2	9	min	30.0	0.3	0.05	0.013
PHI01	Circadian-Rhythms	3	17	h	30.0	0.1	0.1	0.013
GUY02	Cardiovascular-Circulation	2	11	min	30.0	0.3	0.05	0.013
PUL01	Cardiovascular-Circulation	2	561	min	30.0	0.3	0.05	0.013
CAL01	Cell-Cycle	17	43	min	30.0	0.5	0.1	0.013
WOD01	Immunology	3	7	s	30.0	0.1	0.1	0.012
GUP02	Endocrine	4	7	h	30.0	0.5	0.1	0.012
M01	Cardiovascular-Circulation	2	558	min	30.0	0.3	0.05	0.012
LEN02	Endocrine	3	12	s	30.0	0.5	0.1	0.012
KAR02	Signal-Transduction	13	15	s	3.3	0.1	0.1	0.011
SHO02	Excitation-contraction-Coupling	56	105	ms	10.0	0.3	0.05	0.011
MAC01	Ion-Transport	7	35	s	0.33	0.5	0.05	0.01
IRI01	Electrophysiology	23	69	s	0.33	0.3	0.1	0.01
BAG01	Signal-Transduction	9	30	s	3.3	0.1	0.05	0.01
WOL03	Metabolism	13	28	min	30.0	0.5	0.1	0.008
WAN01	Calcium-Dynamics	5	27	s	30.0	0.3	0.1	0.008
NEL01	Immunology	4	7	day	30.0	0.1	0.3	0.007
HUA01	Signal-Transduction	18	37	s	30.0	0.1	0.3	0.007
SVE01	Cell-Cycle	16	58	min	10.0	0.5	0.3	0.007
GUY03	Cardiovascular-Circulation	2	8	min	30.0	0.5	0.05	0.006
CIL01	Cell-Cycle	19	69	min	10.0	0.3	0.05	0.006
LEL01	Signal-Transduction	10	38	h	30.0	0.5	0.05	0.006
GAR01	Cell-Cycle	5	20	min	30.0	0.5	0.3	0.006
COU01	Electrophysiology	21	49	ms	30.0	0.5	0.05	0.005
AUT01	Cardiovascular-Circulation	2	570	min	3.3	0.5	0.05	0.005
GUY04	Cardiovascular-Circulation	2	41	min	3.3	0.5	0.05	0.005
RAZ01	Myofilament-Mechanics	3	19	s	30.0	0.1	0.1	0.005
GRA01	Pkpd	9	30	h	10.0	0.5	0.05	0.005
SNE01	Calcium-Dynamics	5	26	s	1.0	0.5	0.3	0.005
WOD02	Immunology	10	14	day	10.0	0.3	0.1	0.005
WOD03	Immunology	4	12	day	30.0	0.5	0.1	0.004
NEU01	Immunology	3	8	day	10.0	0.3	0.3	0.004
FAR01	Signal-Transduction	5	35	h	1.0	0.1	0.1	0.004
SNE02	Calcium-Dynamics	3	13	s	30.0	0.3	0.1	0.004
FAR02	Signal-Transduction	5	35	h	1.0	0.1	0.1	0.004

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Model	Domain	Channel	Constants	Time Unit	T	d-state	d-const	ATVL
BER02	Electrophysiology	7	47	ms	30.0	0.5	0.1	0.004
GUY05	Cardiovascular-Circulation	3	13	min	30.0	0.3	0.05	0.004
MCA01	Electrophysiology	10	13	ms	30.0	0.3	0.3	0.004
MCA02	Electrophysiology	10	13	ms	30.0	0.3	0.3	0.004
NOV01	Cell-Cycle	11	45	min	30.0	0.5	0.3	0.004
YAM01	Excitation-contraction-Coupling	5	20	s	3.3	0.1	0.1	0.004
RAL01	Metabolism	24	142	min	30.0	0.1	0.3	0.004
VIS01	Electrophysiology	25	75	ms	10.0	0.5	0.05	0.004
CAM01	Myofilament-Mechanics	4	10	s	30.0	0.3	0.05	0.004
VIS02	Electrophysiology	25	75	ms	10.0	0.5	0.05	0.004
SNY01	Calcium-Dynamics	10	27	s	30.0	0.5	0.3	0.003
NOB01	Electrophysiology	20	69	s	0.33	0.1	0.05	0.003
LEL02	Signal-Transduction	3	10	h	30.0	0.5	0.3	0.003
MAL02	Electrophysiology	29	86	ms	30.0	0.5	0.05	0.003
UED01	Signal-Transduction	10	55	h	30.0	0.5	0.1	0.002
CHE01	Cell-Cycle	36	136	min	10.0	0.5	0.3	0.002
POT01	Endocrine	33	97	h	30.0	0.1	0.1	0.002
CHE02	Cell-Cycle	13	71	min	30.0	0.1	0.05	0.002
PAR01	Ion-Transport	7	35	s	3.3	0.5	0.05	0.002
JEL03	Endocrine	2	15	-	30.0	0.1	0.1	0.002
GOO01	Circadian-Rhythms	2	6	s	30.0	0.5	0.3	0.002
DIX01	Immunology	5	18	day	3.3	0.5	0.05	0.002
CSI01	Cell-Cycle	14	93	min	10.0	0.5	0.1	0.002
HAT01	Signal-Transduction	33	80	s	30.0	0.5	0.1	0.002
VAS01	Metabolism	10	19	s	30.0	0.1	0.3	0.002
MIT02	Electrophysiology	2	10	ms	30.0	0.3	0.3	0.002
MIJ01	Myofilament-Mechanics	4	12	s	30.0	0.5	0.3	0.002
REI01	Signal-Transduction	4	11	s	1.0	0.1	0.3	0.002
GUY06	Cardiovascular-Circulation	5	45	min	3.3	0.3	0.05	0.002
GAL01	Electrophysiology	3	28	ms	30.0	0.5	0.05	0.002
LEM01	Endocrine	3	28	day	30.0	0.5	0.05	0.002
SRI01	Cell-Cycle	6	25	min	30.0	0.5	0.05	0.001
FAB01	Electrophysiology	25	81	ms	10.0	0.3	0.05	0.001
CUI01	Calcium-Dynamics	4	22	min	3.3	0.1	0.3	0.001
LEM02	Endocrine	3	28	day	30.0	0.5	0.05	0.001
SCH01	Immunology	5	15	day	30.0	0.5	0.3	0.001
HAU01	Endocrine	4	14	min	30.0	0.5	0.1	0.001
BOR02	Calcium-Dynamics	3	15	min	0.33	0.3	0.05	0.001
WOD04	Immunology	2	5	s	30.0	0.5	0.3	0.001
FIN01	Electrophysiology	27	61	ms	0.33	0.1	0.3	0.001
CHA01	Metabolism	12	43	min	0.33	0.1	0.3	0.001
HAL01	Cell-Cycle	4	7	month	30.0	0.1	0.05	0.001
BAR01	Cell-Cycle	34	55	min	3.3	0.1	0.05	0.001
RAP01	Calcium-Dynamics	11	57	h	10.0	0.5	0.05	0.001
KOM01	Immunology	2	9	h	30.0	0.5	0.1	0.001
NOV02	Cell-Cycle	9	31	min	10.0	0.5	0.05	0.001
NOV03	Cell-Cycle	13	43	min	10.0	0.5	0.05	0.001
VEM01	Signal-Transduction	17	33	s	30.0	0.5	0.3	0.001
MIF01	Electrophysiology	5	14	ms	0.33	0.1	0.3	0.001
COR01	Electrophysiology	14	51	-	30.0	0.5	0.05	0.001
BON01	Immunology	3	7	day	30.0	0.5	0.3	0.001
YAT01	Cell-Cycle	2	5	day	30.0	0.5	0.05	0.001
BON02	Immunology	4	13	day	30.0	0.5	0.3	0.001
BON03	Immunology	3	8	day	30.0	0.5	0.3	0.001
BRO01	Endocrine	2	11	min	30.0	0.5	0.3	0.0
IZA01	Excitation-contraction-Coupling	2	30	ms	10.0	0.1	0.3	0.0
VIL02	Signal-Transduction	6	9	min	30.0	0.5	0.05	0.0
BER03	Electrophysiology	4	21	ms	30.0	0.1	0.3	0.0
BER04	Electrophysiology	4	21	ms	30.0	0.1	0.3	0.0
BER05	Electrophysiology	5	32	ms	30.0	0.5	0.05	0.0
KOM02	Endocrine	3	13	day	30.0	0.1	0.1	0.0
LEN03	Endocrine	5	15	s	30.0	0.5	0.3	0.0
RAT01	Endocrine	3	13	-	30.0	0.5	0.1	0.0
BUE01	Electrophysiology	4	32	ms	30.0	0.5	0.1	0.0
GUY07	Cardiovascular-Circulation	4	27	min	30.0	0.5	0.05	0.0
HUN01	Electrophysiology	29	70	ms	1.0	0.1	0.1	0.0
ROM01	Cell-Cycle	6	28	min	30.0	0.5	0.1	0.0
SHO03	Calcium-Dynamics	7	40	ms	1.0	0.1	0.1	0.0
GUY08	Cardiovascular-Circulation	2	17	min	30.0	0.5	0.05	0.0
POT02	Endocrine	5	12	s	30.0	0.5	0.05	0.0
CIR01	Cardiovascular-Circulation	44	96	ms	30.0	0.1	0.05	0.0
STE01	Excitation-contraction-Coupling	4	5	ms	30.0	0.5	0.3	0.0
PED01	Endocrine	9	54	ms	10.0	0.3	0.05	0.0
GUY09	Cardiovascular-Circulation	2	5	min	30.0	0.1	0.3	0.0
GON01	Endocrine	3	11	min	0.33	0.1	0.05	0.0
ELE01	Cardiovascular-Circulation	4	566	min	30.0	0.1	0.05	0.0
GUY10	Cardiovascular-Circulation	4	17	min	30.0	0.1	0.3	0.0

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Model	Domain	Channel	Constants	Time Unit	T	d-state	d-const	ATVL
POT03	Endocrine	5	16	s	30.0	0.1	0.3	0.0
CLO01	Metabolism	7	42	s	0.33	0.1	0.1	0.0
CLO02	Metabolism	7	42	s	0.33	0.1	0.1	0.0
RAN01	Signal-Transduction	32	32	s	30.0	0.1	0.3	0.0
HYP01	Cardiovascular-Circulation	3	560	min	30.0	0.1	0.3	0.0
GUY11	Cardiovascular-Circulation	3	10	min	30.0	0.1	0.3	0.0
SED01	Signal-Transduction	20	33	min	-	-	-	-
LIV01	Electrophysiology	18	68	ms	-	-	-	-
KAT01	Electrophysiology	9	54	s	-	-	-	-
CUI02	Electrophysiology	20	17	s	-	-	-	-
NIE01	Electrophysiology	26	188	ms	-	-	-	-
CHA02	Electrophysiology	12	38	s	-	-	-	-
KAT02	Electrophysiology	9	54	ms	-	-	-	-
FAV01	Electrophysiology	80	770	s	-	-	-	-
CHA03	Electrophysiology	9	27	s	-	-	-	-
CLO03	Neurobiology	36	125	s	-	-	-	-
VIN01	Metabolism	25	342	min	-	-	-	-
DEM01	Signal-Transduction	29	55	s	-	-	-	-
CHE03	Signal-Transduction	14	20	s	-	-	-	-
COO01	Signal-Transduction	13	38	s	-	-	-	-
NAZ01	Metabolism	8	55	-	-	-	-	-
HUN02	Electrophysiology	29	69	ms	-	-	-	-
NAZ02	Metabolism	8	25	s	-	-	-	-
WAN02	Signal-Transduction	4	4	s	-	-	-	-
CUR01	Metabolism	0	21		-	-	-	-
DAS01	Metabolism	0	30		-	-	-	-
VIN02	Metabolism	25	342	min	-	-	-	-
HEY01	Metabolism	11	18	s	-	-	-	-
CHE04	Signal-Transduction	14	20	s	-	-	-	-
PRO01	Signal-Transduction	14	24	s	-	-	-	-
TRA01	Metabolism	11	71	ms	-	-	-	-
BEA01	Metabolism	19	60	s	-	-	-	-
BEN01	Electrophysiology	29	69	ms	-	-	-	-
NOB02	Electrophysiology	22	70	s	-	-	-	-
LI03	Electrophysiology	12	37	s	-	-	-	-
THA01	Pkpd	2	15	week	-	-	-	-
MAC02	Endocrine	4	14	min	-	-	-	-
MAI01	Cardiovascular-Circulation	7	58	s	-	-	-	-
GUY12	Cardiovascular-Circulation	43	210	min	-	-	-	-
MAI02	Cardiovascular-Circulation	3	21	s	-	-	-	-
KYR01	Endocrine	5	24	min	-	-	-	-
MAC03	Endocrine	7	28	min	-	-	-	-
BAY01	Calcium-Dynamics	3	2	s	-	-	-	-
RIC01	Signal-Transduction	8	11	s	-	-	-	-
BEN02	Mechanical-Constitutive-Laws	7	26	s	-	-	-	-
PAN01	Excitation-contraction-Coupling	22	93	ms	-	-	-	-
MAI03	Cardiovascular-Circulation	7	32	s	-	-	-	-
OVE01	Pkpd	3	37	h	-	-	-	-
PMR01	Gene-Regulation	0	0		-	-	-	-
GUY13	Cardiovascular-Circulation	0	10		-	-	-	-
MOD01	Cardiovascular-Circulation	14	88	s	-	-	-	-
PMR02	Gene-Regulation	0	0		-	-	-	-
GUY14	Cardiovascular-Circulation	4	72	min	-	-	-	-
ESP01	Electrophysiology	21	70	s	-	-	-	-
BEN03	Electrophysiology	29	69	ms	-	-	-	-
BOY01	Electrophysiology	22	90	s	-	-	-	-
ESP02	Electrophysiology	21	70	s	-	-	-	-
ASL02	Electrophysiology	29	49	s	-	-	-	-
OST01	Electrophysiology	8	31	s	-	-	-	-
MIC01	Electrophysiology	43	97	s	-	-	-	-
NIE02	Electrophysiology	5	25	ms	-	-	-	-
INA02	Electrophysiology	29	63	s	-	-	-	-
MAH01	Electrophysiology	26	80	ms	-	-	-	-
SED02	Signal-Transduction	21	38	min	-	-	-	-

E LINKS OF ODE MODELS

Table 5: Model abbreviations with links to their Physiome page. The Physiome page contains information about the ODE model, the respective publication. Furthermore, the generated Python code can also be found there.

Abbreviation	Link
ASL01	https://models.physiomeproject.org/exposure/bb75b3021730966e2827967a66188cb2
ASL02	https://models.physiomeproject.org/exposure/0e3a603db8f464ae89becb8d89225d90
AUT01	https://models.physiomeproject.org/exposure/827af05888f8e152f448d9cd8c6a8d09
BAG01	https://models.physiomeproject.org/exposure/98c3ef8d8c8f444d3a492fe777b4f6d0
BAR01	https://models.physiomeproject.org/exposure/455155cfef38300290a75bcfe7440c31
BAY01	https://models.physiomeproject.org/exposure/210f6601f6461be8443592ff071d2592
BEA01	https://models.physiomeproject.org/exposure/959a4da5d8a0638f4b36adf5800f4fc0
BEN01	https://models.physiomeproject.org/exposure/5d2fd39aa16f562d3bc19119847b7db0
BEN02	https://models.physiomeproject.org/exposure/b503501533abcf0e70786789f08cb902
BEN03	https://models.physiomeproject.org/exposure/5d2fd39aa16f562d3bc19119847b7db0
BER01	https://models.physiomeproject.org/exposure/c6d3b19b455fbb3f80813337ac127674
BER02	https://models.physiomeproject.org/exposure/a7f8b4574fe4802f1e06f247006962ac
BER03	https://models.physiomeproject.org/exposure/fe91f06c8311ffd90fd1eaabe9b0032a
BER04	https://models.physiomeproject.org/exposure/fe91f06c8311ffd90fd1eaabe9b0032a
BER05	https://models.physiomeproject.org/exposure/48e767ee37f347e1a3876d0549a0866b
BON01	https://models.physiomeproject.org/exposure/def2cd06391e17d8966e22997a65476a
BON02	https://models.physiomeproject.org/exposure/def2cd06391e17d8966e22997a65476a
BON03	https://models.physiomeproject.org/exposure/def2cd06391e17d8966e22997a65476a
BOR01	https://models.physiomeproject.org/exposure/7ba127c00a5524c49f0a57e6589eda03
BOR02	https://models.physiomeproject.org/exposure/7ba127c00a5524c49f0a57e6589eda03
BOY01	https://models.physiomeproject.org/exposure/d4d2febdd2a9556eca88ceeac66bf696
BRO01	https://models.physiomeproject.org/exposure/1f40088c3eb67895ff8dcc5010d62bf1
BUE01	https://models.physiomeproject.org/exposure/954421556db2a0c7de89ce0bdc26bed9
BUT01	https://models.physiomeproject.org/exposure/293a909eeeca07d0ca8ad583839eb0bc
BUT02	https://models.physiomeproject.org/exposure/293a909eeeca07d0ca8ad583839eb0bc
CAL01	https://models.physiomeproject.org/exposure/1a3f36d015121d5596565fe7d9afb332
CAM01	https://models.physiomeproject.org/exposure/62183706711e435ff002b46088540850
CHA01	https://models.physiomeproject.org/exposure/a74fd808df7ba454945e5ac42486e3ad
CHA02	https://models.physiomeproject.org/exposure/1f4c6ffdb5120ed09675e4c9d346952f
CHA03	https://models.physiomeproject.org/exposure/7fce899159b8698ccb2c8569d19a0ea1
CHE01	https://models.physiomeproject.org/exposure/0e2337ffa4c550dd077c629b9e76330e
CHE02	https://models.physiomeproject.org/exposure/635c1359f216ab8d064dfe8e8d3387b8
CHE03	https://models.physiomeproject.org/exposure/01f8729559c3985a3d09ff4d983da75a
CHE04	https://models.physiomeproject.org/exposure/01f8729559c3985a3d09ff4d983da75a
CIL01	https://models.physiomeproject.org/exposure/f5be28698749563c5564d1a4fa5c0446
CIR01	https://models.physiomeproject.org/exposure/ff8be5f140e68612284488cf9879eb5f
CLO01	https://models.physiomeproject.org/exposure/e5cfb42225d4534a1e08979e57cf8bdd
CLO02	https://models.physiomeproject.org/exposure/e5cfb42225d4534a1e08979e57cf8bdd
CLO03	https://models.physiomeproject.org/exposure/6e249a04f5c751e42ba41504d84e6e49
COO01	https://models.physiomeproject.org/exposure/39fee05138f428413db61aa29ff6f0de
COR01	https://models.physiomeproject.org/exposure/9477b42c85e81f7da21059baa509ae21
COU01	https://models.physiomeproject.org/exposure/0e03bbe01606be5811691f9d5de10b65
CSI01	https://models.physiomeproject.org/exposure/dada9c9d39d0783375d42f12f11dddf3
CUI01	https://models.physiomeproject.org/exposure/a3fb8998a3dd86ca914e55d60b3b7a52
CUI02	https://models.physiomeproject.org/exposure/45ddea4fcde109d7a2bef806aeae84d7
CUR01	https://models.physiomeproject.org/exposure/e08bd642df06eb59404c8e46afbb04a4
DAS01	https://models.physiomeproject.org/exposure/bb0ee0e6d141a529fa896ff3785f0eff
DEM01	https://models.physiomeproject.org/exposure/32c9e9739454b40b5ba2d9cabb1fd079
DIF01	https://models.physiomeproject.org/exposure/91d93b61d7da56b6baf1f0c4d88ecd77
DIX01	https://models.physiomeproject.org/exposure/cb02cc9ce006eb44b2077111c1d548d2
DOK01	https://models.physiomeproject.org/exposure/462ab10275dfc099166c8a0e4f9e1be3
DUP01	https://models.physiomeproject.org/exposure/060c119cc5365e3d9cd0203c82fe0121
DUP02	https://models.physiomeproject.org/exposure/04cc3dc1c0c8e5dc3634f0220904d297
DUP03	https://models.physiomeproject.org/exposure/060c119cc5365e3d9cd0203c82fe0121
ELE01	https://models.physiomeproject.org/exposure/239e0bdec0c7054e0727fc111a54658c
ESP01	https://models.physiomeproject.org/exposure/3ead51a9899108b4f8c8177c0b5f2421
ESP02	https://models.physiomeproject.org/exposure/3ead51a9899108b4f8c8177c0b5f2421
FAB01	https://models.physiomeproject.org/exposure/55643f2114a2a463ada007deb9fc3913
FAR01	https://models.physiomeproject.org/exposure/fe9ed942c523777274fcdb43c6be75f7
FAR02	https://models.physiomeproject.org/exposure/fe9ed942c523777274fcdb43c6be75f7
FAV01	https://models.physiomeproject.org/exposure/f17a3cade0f0f7986b060b8bbf5c2957
FIN01	https://models.physiomeproject.org/exposure/eeb81adc372c2f172399ec7160b0331e
GAL01	https://models.physiomeproject.org/exposure/bc0d80f7e1ccb2c288119f6e8fdfbc20
GAR01	https://models.physiomeproject.org/exposure/4f5fb67ce413238e24e5277f46fd5d21
GON01	https://models.physiomeproject.org/exposure/07464e63da82057cc6e44d33ec60cd7c
GOO01	https://models.physiomeproject.org/exposure/b69b4d821f5b2996cda5f2b4a3e9ca8f
GRA01	https://models.physiomeproject.org/exposure/32d3dcc3ab074d17905c10f2ba26b54f
GUP01	https://models.physiomeproject.org/exposure/9b91dc35df3c4ca3ee14f6a3f4191db7

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Abbreviation	Link
GUP02	https://models.physiomeproject.org/exposure/9b91dc35df3c4ca3ee14f6a3f4191db7
GUY01	https://models.physiomeproject.org/exposure/d20c1a1db1b4027605b87b4361e62560
GUY02	https://models.physiomeproject.org/exposure/63f124c018745aa8ec92a091434f3dfc
GUY03	https://models.physiomeproject.org/exposure/c1461e469a5e282face2fdf37817ca10
GUY04	https://models.physiomeproject.org/exposure/827af05888f8e152f448d9cd8c6a8d09
GUY05	https://models.physiomeproject.org/exposure/be69ef6c1a42264162ae6e370e6107b5
GUY06	https://models.physiomeproject.org/exposure/e1f152d0dbd1dec4111faa117db85bc0
GUY07	https://models.physiomeproject.org/exposure/f3272a51c6e95c70eb1309d30c08d4cf
GUY08	https://models.physiomeproject.org/exposure/f97a5eb092b12f4f0f32ac51ee20d20e
GUY09	https://models.physiomeproject.org/exposure/64f53aa4552f15dc6805444ba0559c5a
GUY10	https://models.physiomeproject.org/exposure/239e0bdec0c7054e0727fc111a54658c
GUY11	https://models.physiomeproject.org/exposure/ea5144be46b69ba503a2d0cd5cbd5b96
GUY12	https://models.physiomeproject.org/exposure/cd10322c000e6ff64441464f8773ed83
GUY13	https://models.physiomeproject.org/exposure/4c73baa30cb81cf0cd83b4e8d42a5358
GUY14	https://models.physiomeproject.org/exposure/b897ad1b96031d40293b2e2e10684ffe
HAL01	https://models.physiomeproject.org/exposure/77bdebf1280135c36f94cac077f74e18
HAT01	https://models.physiomeproject.org/exposure/c14e77a831ec3f1e56ef03240986a8b7
HAU01	https://models.physiomeproject.org/exposure/f9b96930ad2e7f5c9e15b1169a9378c7
HEY01	https://models.physiomeproject.org/exposure/3c875389313cd73f120d69a59cc65acb
HOD01	https://models.physiomeproject.org/exposure/5d116522c3b43ccaeb87a1ed10139016
HUA01	https://models.physiomeproject.org/exposure/3227298fc9cc3afc14d058750142c69c
HUN01	https://models.physiomeproject.org/exposure/f4b7120aa512c7f5e7a0664abcee3e8b
HUN02	https://models.physiomeproject.org/exposure/f4b7120aa512c7f5e7a0664abcee3e8b
HYN01	https://models.physiomeproject.org/exposure/b5c9ac7a7a8b76918c02b39967ae191b
HYP01	https://models.physiomeproject.org/exposure/ea5144be46b69ba503a2d0cd5cbd5b96
INA01	https://models.physiomeproject.org/exposure/08bcead2dc05cf2709a598e7f61a6182
INA02	https://models.physiomeproject.org/exposure/08bcead2dc05cf2709a598e7f61a6182
IRI01	https://models.physiomeproject.org/exposure/7af88a567b3fdbb5209ddd2eee44075c
IZA01	https://models.physiomeproject.org/exposure/94a5dddfa94d46f1d842e4e87a94646f
JEL01	https://models.physiomeproject.org/exposure/4817bdfdd245412f52dd79da59137fe8
JEL02	https://models.physiomeproject.org/exposure/4817bdfdd245412f52dd79da59137fe8
JEL03	https://models.physiomeproject.org/exposure/4817bdfdd245412f52dd79da59137fe8
KAR01	https://models.physiomeproject.org/exposure/b5ddf62f58c911e794174d61f9aedf61
KAR02	https://models.physiomeproject.org/exposure/46a466d519479cee832ebf8b66c1ea00
KAT01	https://models.physiomeproject.org/exposure/2a3e1e865606c4da94c251017038f820
KAT02	https://models.physiomeproject.org/exposure/2a3e1e865606c4da94c251017038f820
KOM01	https://models.physiomeproject.org/exposure/88ec4bc9780f123780b75e4c9ce1682b
KOM02	https://models.physiomeproject.org/exposure/fa7996b256f19f92568fb7ad414b6cba
KYR01	https://models.physiomeproject.org/exposure/fd3eeaae0cbb662a9d5a2b47ce81a825
LEL01	https://models.physiomeproject.org/exposure/1db3a968ce8585201acd753ce857002a
LEL02	https://models.physiomeproject.org/exposure/1db3a968ce8585201acd753ce857002a
LEM01	https://models.physiomeproject.org/exposure/3094a5c3e0810028a9accc3b773cab36
LEM02	https://models.physiomeproject.org/exposure/3094a5c3e0810028a9accc3b773cab36
LEN01	https://models.physiomeproject.org/exposure/fb612e6ce429bc45d99388c045a100cb
LEN02	https://models.physiomeproject.org/exposure/520cbff71e195d0a0ed36ce1c78d46d5
LEN03	https://models.physiomeproject.org/exposure/520cbff71e195d0a0ed36ce1c78d46d5
LI01	https://models.physiomeproject.org/exposure/dfe4f6c90d58266f0f5d6d320c291e40
LI02	https://models.physiomeproject.org/exposure/dfe4f6c90d58266f0f5d6d320c291e40
LI03	https://models.physiomeproject.org/exposure/dfe4f6c90d58266f0f5d6d320c291e40
LIV01	https://models.physiomeproject.org/exposure/dedfcd1a135ddb59e9d979ec1376a44f
M01	https://models.physiomeproject.org/exposure/d20c1a1db1b4027605b87b4361e62560
MAC01	https://models.physiomeproject.org/exposure/a499a7082706164315ad07feff408850
MAC02	https://models.physiomeproject.org/exposure/1229d793e4a010088736d682bd35d463
MAC03	https://models.physiomeproject.org/exposure/1229d793e4a010088736d682bd35d463
MAH01	https://models.physiomeproject.org/exposure/a5586b72d07ce03fc40fc98ee846d7a5
MAI01	https://models.physiomeproject.org/exposure/a195d957a2d63ac4defe3232fd0ea50c
MAI02	https://models.physiomeproject.org/exposure/9970597960dbe3a381d773d90c0298c2
MAI03	https://models.physiomeproject.org/exposure/feccb0f0649425a61526049f35c3b7fb
MAL01	https://models.physiomeproject.org/exposure/8e081783cbe18d346eaf14cd5c1ae18f
MAL02	https://models.physiomeproject.org/exposure/3c1c7b17df06921a4e1b05c639a45d32
MCA01	https://models.physiomeproject.org/exposure/60e23c3228a3e455699846704006a8fe
MCA02	https://models.physiomeproject.org/exposure/60e23c3228a3e455699846704006a8fe
MIC01	https://models.physiomeproject.org/exposure/9e77fb28778473706ebd43bd5bea86d7
MIF01	https://models.physiomeproject.org/exposure/8854e495f7e4240b1607820088b13138
MIJ01	https://models.physiomeproject.org/exposure/a9cd3fd1ec3c3bbe2613ca9e7490a367
MIT01	https://models.physiomeproject.org/exposure/0d32bf39c8af51c44373b2e68c3cec74
MIT02	https://models.physiomeproject.org/exposure/9efc295a6f3360044b4712c28b8314e7
MOD01	https://models.physiomeproject.org/exposure/c49d416ae3a5132882e6ea7479ba50f5
NAZ01	https://models.physiomeproject.org/exposure/45de427094d4814e74156890fd99fcc6
NAZ02	https://models.physiomeproject.org/exposure/45de427094d4814e74156890fd99fcc6
NEL01	https://models.physiomeproject.org/exposure/e811b8311a550fb0eedf95402b3166d0
NEU01	https://models.physiomeproject.org/exposure/a7e3d6ea01b67ecbe06e11f7909bcb93
NIE01	https://models.physiomeproject.org/exposure/59a44249dec83576d97fd3fce46ec5f9
NIE02	https://models.physiomeproject.org/exposure/97fb1de5199b1a74c89281db97aecc13
NOB01	https://models.physiomeproject.org/exposure/e42fb39e149f444be872e04e6e731ac0
NOB02	https://models.physiomeproject.org/exposure/a40c4434423c0436e2789a2d457b7ab2
NOV01	https://models.physiomeproject.org/exposure/ea5f37b01b27c7ba25b6c179a12a464a

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Abbreviation	Link
NOV02	https://models.physiomeproject.org/exposure/1e1bee6ef3243503e7e1531cfd61bb3f
NOV03	https://models.physiomeproject.org/exposure/e24887f982e9246d05ba0f7152bd4aaa
NYG01	https://models.physiomeproject.org/exposure/ad761ce160f3b4077bbae7a004c229e3
OST01	https://models.physiomeproject.org/exposure/d9de93b128da322a4d50f24589980ea1
OVE01	https://models.physiomeproject.org/exposure/504e7681708fc7b1260db25363658be5
PAR01	https://models.physiomeproject.org/exposure/a07c660076d3caddac2d136b0c0c7c05
PED01	https://models.physiomeproject.org/exposure/00bc5ddf4f576af8f81ffc429a183a73
PHI01	https://models.physiomeproject.org/exposure/87ef0e70902ed8a49d10252617835aa9
PMR01	https://models.physiomeproject.org/exposure/f2a5a551e8e22f18e5adb641243269c5
PMR01	https://models.physiomeproject.org/exposure/62b7b4336d5d1caa0a01c3faff8b33ea
PMR01	https://models.physiomeproject.org/exposure/2fa5de12b0923874aee04486617eb2bd
PMR02	https://models.physiomeproject.org/exposure/52e2a146e8bf5976379803fec46138a7
PMR02	https://models.physiomeproject.org/exposure/29a0ec2468a49a64a123f927083260f0
POT01	https://models.physiomeproject.org/exposure/05510be013e1d096e26c3716c950712d
POT02	https://models.physiomeproject.org/exposure/80982c99e643a576d10e7f3e271a2299
POT03	https://models.physiomeproject.org/exposure/80982c99e643a576d10e7f3e271a2299
PRO01	https://models.physiomeproject.org/exposure/026ccefa76a5ea437d6339aae90c2d37
PUL01	https://models.physiomeproject.org/exposure/63f124c018745aa8ec92a091434f3dfc
PUR01	https://models.physiomeproject.org/exposure/baa78a664bd1f620acce7257883fc91a
PUR02	https://models.physiomeproject.org/exposure/baa78a664bd1f620acce7257883fc91a
RAL01	https://models.physiomeproject.org/exposure/d8dae90ac67ab8b892864c7c66bf6110
RAN01	https://models.physiomeproject.org/exposure/9fb32c29b3650b853590d30c7e3374af
RAP01	https://models.physiomeproject.org/exposure/ea84bd6e27e521677bfc2d6283b371f3
RAT01	https://models.physiomeproject.org/exposure/f9611ad158302284299c1020faf878ee
RAZ01	https://models.physiomeproject.org/exposure/6703dcc072ed1c21cade028b9abbcfa0
REE01	https://models.physiomeproject.org/exposure/b0a3d0dbb029b6cbd2a193014b779a82
REI01	https://models.physiomeproject.org/exposure/cfbc61b251bcccb085ffd9d074c1d44c
REV01	https://models.physiomeproject.org/exposure/03852e75c08161cf63d70417ec4a2fae
RIC01	https://models.physiomeproject.org/exposure/e340f005288ec6bd49535cf792769dd0
ROM01	https://models.physiomeproject.org/exposure/9a75ea45513cec9ac59991afa40430a4
SCH01	https://models.physiomeproject.org/exposure/926ad168d96e3fe4b87f452890631f86
SED01	https://models.physiomeproject.org/exposure/c18a94e82ef7ed7c9dad399e61677b5e
SED02	https://models.physiomeproject.org/exposure/c18a94e82ef7ed7c9dad399e61677b5e
SHO03	https://models.physiomeproject.org/exposure/e10def01073e56fb32f06bf59e66a318
SNE01	https://models.physiomeproject.org/exposure/4f67bf3d82a1ad07d332e383937041e2
SNE02	https://models.physiomeproject.org/exposure/88053fd5c8867b72ee23c1702097e377
SNY01	https://models.physiomeproject.org/exposure/8d791fd6aba9793fb0149d5b6e3afcee
SRI01	https://models.physiomeproject.org/exposure/500b72e21302febd0f3dc746dc22af81
STE01	https://models.physiomeproject.org/exposure/b060fdbcfae8c7d85e595c24d36ab11b
SVE01	https://models.physiomeproject.org/exposure/aad2ca9cce58dba12c8d9d85dccada0f
THA01	https://models.physiomeproject.org/exposure/1985e7c820ff102b1caebc241df65ce7
TRA01	https://models.physiomeproject.org/exposure/cfa7684fb084ad748bf3061569d99334
UED01	https://models.physiomeproject.org/exposure/738882ee938d6b2e2d264714259e0416
VAN01	https://models.physiomeproject.org/exposure/a0df3e434c6d37073783883cb4dff2d3
VAS01	https://models.physiomeproject.org/exposure/05557c5c5ae347b0a6099f66eb33e50f
VEM01	https://models.physiomeproject.org/exposure/c1e5f2110188582f6f44f85930733144
VIL01	https://models.physiomeproject.org/exposure/d5af7e23b0b67c042ca76949c7e59c79
VIL02	https://models.physiomeproject.org/exposure/882bd564f637b7a506fafdf8a953a506
VIN01	https://models.physiomeproject.org/exposure/07cdb663179c77de23e425a56464a2aa
VIN02	https://models.physiomeproject.org/exposure/07cdb663179c77de23e425a56464a2aa
VIS01	https://models.physiomeproject.org/exposure/4c26896c035fd04e70658d792affad9d
VIS02	https://models.physiomeproject.org/exposure/4c26896c035fd04e70658d792affad9d
WAN01	https://models.physiomeproject.org/exposure/a9e94f2a37908bf3ebb1d6aa4bef82c4
WAN02	https://models.physiomeproject.org/exposure/a93163783c79b766b9ab88c8a20c1e41
WOD01	https://models.physiomeproject.org/exposure/eb9280eef64afce83878426e24d5b31c
WOD02	https://models.physiomeproject.org/exposure/9fddc0e3b1bb1d6ad86baeea2e053715
WOD03	https://models.physiomeproject.org/exposure/869cfba0248523dcd57a7e49185f70ab
WOD04	https://models.physiomeproject.org/exposure/eb9280eef64afce83878426e24d5b31c
WOL01	https://models.physiomeproject.org/exposure/c7801370ba2f7c3e19ad4d12d55d4968
WOL02	https://models.physiomeproject.org/exposure/c1821457cc8c8135f3317ec4882b719f
WOL03	https://models.physiomeproject.org/exposure/f68c97d153bb0e4a050cb4e143b047a8
YAM01	https://models.physiomeproject.org/exposure/5ad3bc65fe5ed254806310e2dfb1c5cb
YAT01	https://models.physiomeproject.org/exposure/0b98c29503e70a6b35be0ed0f5719458

F Hyperparameter Search

We exclusively conduct hyperparameter search on fold 0. For **GraFITi** (Yalavarthi et al., 2024) the hyperparameters for the search are as follows:

- The number of layers, with possible values [1, 2, 3, 4].
- The number of attention heads, with possible values [1, 2, 4].
- The latent dimension, with possible values [16, 32, 64, 128, 256].

For the **LinODEnet** model (Scholz et al., 2022) we search the hyperparameters from:

- The hidden dimension, with possible values [16, 32, 64, 128].
- The latent dimension, with possible values [64, 128, 192, 256].

For **GRU-ODE-Bayes** (De Brouwer et al., 2019) we tune the hidden size from [16, 32, 64, 128, 256]

For **Neural Flows** (Biloš et al., 2021) we define the hyperparameter spaces for the search are as follows:

- The number of flow layers, with possible values [1, 2, 4].
- The hidden dimension, with possible values [16, 32, 64, 128, 256].
- The flow model type, with possible values [GRU, ResNet].

For the **CRU** (Schirmer et al., 2022) the hyperparameter space is as follows:

- The latent state dimension, with possible values [10, 20, 30].
- The number of basis functions, with possible values [10, 20].
- The bandwidth with possible values [3, 10].

G Regularly Sampled Time Series Data Experiments

In this chapter, we show important findings related to generating **regularly sampled multivariate time series data** with `Physiome-ODE`. We discuss the generation process of 3 datasets that were chosen based on 2 important criterions. First is the previously mentioned JGD which was shown to be important indicator for the complexity of the dataset. Second criterion is for these datasets to include a high number of channels as we show through the following small experiment, how channel dependency matters more and have more effect when tested on our ODE-datasets compared to current benchmark datasets.

The datasets used here were DOK01, INA01, and HYN01 which all have a number of channels greater than 15 with 18, 29, and 22 channels respectively. Furthermore, they are ranked highly with respect to the JGD metric having values of 2,277, 2,218, and 1,548 respectively which makes them the most complex high dimensional (more than 15 channels) datasets in our benchmark. Datasets were generated based on the configuration at Appendix D, generating 100 samples for each dataset based on 100 different initial conditions that are drawn uniformly at random from a distribution based on the configuration given to the model. Each time series sample spans 300 timesteps. The datasets were split into a train/val/test split of 70/20/10 which is similar to the split done normally on the TSF literature for the weather, electricity and traffic datasets (Chen et al., 2023; Zeng et al., 2023; Nie et al., 2023).

We conduct experiments on state-of-the-art Time Series Forecasting models PatchTST (Nie et al., 2023), TSMixer (Chen et al., 2023), and DLinear (Zeng et al., 2023). For DLinear, there is only a univariate model (doesn't capture channel dependency), but for the other models, they can be implemented as univariate or multivariate (capture channel dependency) variants. So overall, we label the univariate models as **CI** (**C**hannel **I**ndepedent), and multivariate models as **CD** (**C**hannel **D**ependent). These models are tested with the 3 ODE datasets mentioned earlier tuning the hyperparameters for the 3 models through random search of 20 trials for a predefined range of parameters.

For PatchTST, we set the following hyperparameter range:

- learning rate (range of floats) $\rightarrow$ [10^{-7}, 10^{-2}].
- number of encode layers (range of integers) $\rightarrow$ [1, 6].
- Dimensionality of fully connected layer (categorical) $\rightarrow$ [256, 512, 1024].
- Embedding dimension of the attention layer (categorical) $\rightarrow$ [128, 256, 512, 1024].
- dropout (range of floats) $\rightarrow$ [0, 0.9].
- fully connected layer dropout (range of floats) $\rightarrow$ [0, 0.9].
- patch size (categorical) $\rightarrow$ [4, 8].
- stride (categorical) $\rightarrow$ [2, 4].

For TSMixer, we set the following hyperparameter range:

- learning rate (range of floats) $\rightarrow$ [10^{-7}, 10^{-2}].
- number of blocks (range of integers) $\rightarrow$ [1, 5].
- hidden size of MLP layers (categorical) $\rightarrow$ [32, 64, 256, 1024].
- dropout (range of floats) $\rightarrow$ [0, 0.9].
- activation (categorical) $\rightarrow$ [ReLU, GeLU].

For DLinear, learning rate was tuned on the same range of values mentioned above. All models were tuned on the validation set. The results for the best performing model on validation set (lowest MSE error) were reported on the test set. We used a lookback window of 24 and forecasting horizon of 12 for all the conducted experiments on the ODE datasets. On the other hand, we report the results of the mentioned models on well-established evaluation datasets such as ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity, and Traffic on a forecasting horizon of 96. The reported results from Table 6 give us interesting insights about our benchmark on the regularly sampled time series data.

Table 6: MSE Test loss reported on 3 ODE datasets as well as on 7 of the most popular regular TSF datasets. We highlight the best performing version (CI/CD) of each model in **bold**. The results were reported on 5 different baselines showing a clear trend of CD models being better on our generated ODE datasets. On the other hand, for benchmark datasets, this trend is not there as CI models are performing quite well.

Dataset	PatchTST(CI)	PatchTST(CD)	TSMixer(CI)	TSMixer(CD)	DLinear
INA01	0.099	**0.052**	0.168	**0.062**	0.191
DOK01	0.087	**0.022**	0.155	**0.039**	0178
HYN01	**0.001**	**0.001**	0.019	**0.002**	0.046
ETTh1	**0.375***	0.416*	0.361*	**0.359***	0.375*
ETTh2	**0.274***	0.334*	**0.274***	0.275*	0.289*
ETTm1	**0.290***	0.326*	0.285*	**0.284***	0.299*
ETTm2	**0.165***	0.195*	0.163*	**0.162***	0.167*
Weather	**0.152***	0.168*	**0.145***	**0.145***	0.176*
Electricity	**0.130***	0.196*	**0.131***	0.132*	0.140*
Traffic	**0.367***	0.595*	0.376*	**0.370***	0.410*

[*] Results are taken directly for PatchTST, TSMixer, and DLinear from the respective papers (Nie et al., 2023), (Chen et al., 2023), and (Zeng et al., 2023).

We notice from the results how unpredictable the results can be for the benchmark datasets, with **CI** models being better on some datasets and **CD** models being better on the rest. Also, one more interesting finding is how close the test loss is between a very simple CI model such as DLinear and the state-of-the-art models which can be concerning regarding how much channel dependency there is on the available benchmark datasets. On the other hand, for our ODE-generated datasets, the results are more consistent with clear better performance for **CD** models over their **CI** counterparts. Also, the best model overall was noticed to be always a **CD** model which shows the importance of capturing channel dependency on the ODE generated datasets. One more interesting finding is how worse is a DLinear model on our benchmark datasets where its results are at least 5 times worse than the best performing model. That shows how complex these datasets are, as well as how multivariate models can clearly benefit from intrinsic relation between the channels on such datasets.

H Lorenz

We conducted a forecasting experiment on the chaotic Lorenz Attractor:

$$\frac{dx}{dt} = \sigma(y - x) \tag{40}$$

$$\frac{dy}{dt} = x(\rho - z) - y \tag{41}$$

$$\frac{dz}{dt} = xy - \beta z \tag{42}$$

Instead of varying the constants as we do in `Physiome-ODE`, we set the parameters to

- $\rho = 28$
- $\sigma = 10$
- $\beta = \frac{8}{3}$

We vary the sample the initial states from $x \sim [1, 3], y \sim [0, 2] z \sim [0, 2]$. Following Gilpin (2021), the task is to forecast the final $\frac{1}{6}$ of the IMTS based on the initial $\frac{5}{6}$. Since we always use the exact same constants, we create 200 instead of 2000 time series instances. Outside the mentioned changes, the experimental protocol is the same as the one used in our experiments with the ODEs from `Physiome-ODE`.

As shown in Table 7, there is no model which significantly outperforms the constant baseline GraFITi-C.

Table 7: Test MSE IMTS created on the chaotic Lorenz-Attractor.

Dataset	GRU-ODE	LinODEnet	CRU	Neural Flow	GraFITi	GraFITi-C
Lorenz	1.334±0.097	1.313±0.123	1.292±0.087	1.345±0.105	1.294±0.090	1.303±0.079